

UAS Administration Guide - 6.0.1-GA (AI)

Table of Contents

1. Preface - UAS Administration Guide.....	3
2. Introduction to UAS Administration.....	4
3. Token.....	6
3.1. Token Management.....	7
3.2. Token Administration.....	31
3.3. Report.....	41
3.3.1. System & Admin.....	42
3.3.2. Token & SMS.....	44
4. Administration for End-to-End Encryption Authentication.....	52
5. Administration for Contextual Authentication.....	53
6. Biometrics.....	74
7. Administration for RADIUS.....	77
8. Implement Plugin for Remote Token Server.....	90
9. Appendices.....	94

1. Preface - UAS Administration Guide

This guide provides the concept of AccessMatrix Universal Authentication Server (UAS) and the details about how to configure UAS features.



Note: The term <ACMATRIX_ROOT> refers to the root or installation directory of AccessMatrix on both Windows and Unix platforms.

Intended Reader

This is intended as a reference for system administrators.

Document Scope

This document provides instructions on how to perform administrative tasks using the AccessMatrix UAS product.

For feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

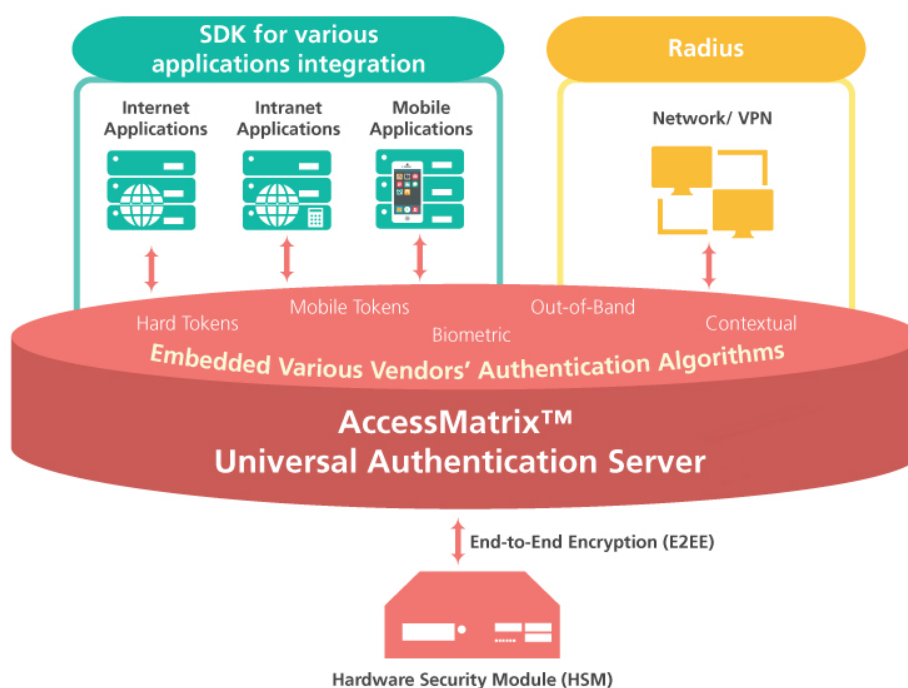
For more information, see the following documents in the AccessMatrix Product Suite:

- [AccessMatrix Common Administration Guide](#)
- [UAS SMS OTP Implementation Guide](#)
- [UAS OneSpan Token Implementation Guide](#)
- [UAS E2EEA FM with SafeNet ProtectServer HSM Implementation Guide](#)
- [UAS E2EEA with SafeNet Payment HSM Implementation Guide](#)
- [UAS Pseudo E2EEA Implementation Guide](#)
- [UAS PDF Pin Mailer Implementation Guide](#)

2. Introduction to UAS Administration

AccessMatrix Universal Authentication Server (UAS) is a comprehensive and versatile authentication server that enable organizations to centralize all their authentication schemes in a single unified platform. It is a future-proof authentication infrastructure that supports multiple authentication mechanisms (such as passwords, OTP, biometrics) for strong authentication and authorization requirements.

UAS offers the freedom for organizations to deploy various authentication methods from different vendors under one platform, and manage it as a holistic security approach. This means organizations will not be restricted or locked to a specific hardware or software authentication solution. The use of a Pluggable Authentication Module (PAM) approach also allows organizations to have the ease of adding new methods when available on top of their existing scheme.



UAS provides security Administration, Authentication, Authorization, and Audit services for business applications. In addition to authentication, it provides:

- End-to-End Life Cycle Management for OTP Tokens.
 - Security services APIs for application integration and shield the complexity of token integration.
 - Radius ready server to integrate with network devices.
 - Administration module for administration delegation and user management.
-

-
- Audit and reporting module.
 - HSM interface to leading brands of HSM devices.
 - PIN mailer interface for mail merge and secure printing.
 - Enforcing password policy - password history, password ageing, password quality check, etc.

Key Benefits

- Reduce operational cost with a common authentication platform.
- No vendor lock-in.
- Simplify integration and deployment efforts.
- Handle complex authentication requirements.
- Cater for future authentication options.
- Provide a highly scalable, open and reliable platform.

Features

- **Versatile Authentication Support**

UAS provides out-of-the-box PAM modules to support various authentication mechanisms like secure passwords, OTP Tokens and Biometric Devices.

- **Complete Token Life Cycle Management & Administration**

UAS provides an integrated solution to manage the lifecycle of OTP token devices, including issuance, delivery, out of sync, lost token, temporary access, replacement over time and other supporting functions such as PIN mailer integration, a reporting module, etc.

- **Embedded Vendor Agnostic Authentication Support for OTP and Biometrics**

UAS provides embedded authentication support for tokens from various vendors like OneSpan (Vasco), DynamiCode, i-Sprint YESsafe Token, Gemalto, V-Key, OATH-based vendors and biometric devices.

- **Comprehensive Support for Mobile Device-based authentication**

UAS offers a choice of mobile token integration to meet different needs: OneSpan (Vasco) DIGIPASS for Mobile, i-Sprint YESsafe OATH-based token and Google authenticator. UAS also provides a self-service soft token activation portal for OneSpan (Vasco) DIGIPASS for Mobile.

- **Extensible to SSO on the same platform**

UAS provides Single Sign-On (SSO) capabilities on the same platform.

3. Token

UAS provides a unified token management platform in support of multiple authentication methods and mechanisms for strong authentication and authorization requirements. It supports embedded authentication of OTP tokens such as i-Sprint SMS, VASCO, OATH-based vendors such as Google Authenticator Software Token and i-Sprint's YESsafe Token. These OTP tokens are often used as a part of multi-factor authentication (MFA). MFA aids in countering hacking attacks and identity fraud.

Supported tokens:

- **Hardware Security Token (e.g. VASCO)** - is a physical device that generates a one-time password to gain access to an electronically restricted resource. This token is used in addition to or in place of a password. See [UAS OneSpan Token Implementation Guide](#) for details.
- **Out-of-Band (OOB) OTP (e.g. i-Sprint SMS)** - this mechanism delivers an OTP to users via short message service (SMS) or email which is different from the channel that the users interact with applications. When a user tries to access a website, an OTP will be sent to the user's mobile phone via SMS or email which has been determined by the user in advance. See [UAS SMS OTP Implementation Guide](#) for details.
- **Open Authentication (OATH) Token (e.g. YESsafe Token)** - is a secure one time password that uses an OATH encryption algorithm, an open-source algorithm that will be used as the base OTP algorithm for soft and hard security tokens.

The generation of OTP will be according to a configurable token policy. The token policy governs how the token used for authentication or authorization behaves. Therefore, the admin must first identify what is the token policy to be used and whether the application needs to supply extra parameters for the generation of OTP. The token admin is required to configure the respective token policy based on the OTP token used.

3.1. Token Management

This section provides details on managing tokens using AccessMatrix UAS.

Token Management Features

Token Initialization

In token initialization:

- The token issuers must decide whether the token is factory initialized or in-house initialized based on the company's own security policy and risk assessment.
- The token issuers must decide on a few token initialization parameters, e.g. length of OTP, crypto-algorithm used (e.g. 3DES or AES), crypto-application Id, etc. For VASCO token, refer to relevant VASCO documentation for the possible parameters.
- For both factory and in-house initialization, an encrypted token seeds file and the encryption key for that file will be the deliverables. Token issuers are advised to have a proper control procedure to manage the token seeds file and the encryption key. For example, different persons may hold the token seeds file and the encryption key.



Note: No UAS component is involved in this process.

Token Activation

For soft token (software-based token), some tokens may require an activation process depending on the soft token vendor. The activation process, in general, is meant to provision the token seed to the device/phone and to verify that the token seed provisioned to the device/phone is correct (normally based on OTP/signature verification). This process normally is user self-service and involves a few steps, which are very dependent on the token vendors.

- OneSpan (formerly VASCO) Multi-Device requires some data to be exchanged between UAS and the OneSpan soft token. It has the options of having online or offline activation. AccessMatrix UAS Token Activation Portal (UAS TAP) module is also available for a self-service token of OneSpan Multi-Device for offline activation.
 - V-Key OATH Token has a separate token seed provisioning server to let the V-Key soft token download the token seed embedded in V-OS. In the activation process, AccessMatrix UAS is used to provide the token serial number and token activation password.
-

-
- YESsafe Token has an activation process that lets the OATH token seed be generated on the fly in UAS.

In term of the token activation process, UAS provides the administration on how to prepare the policy or the token management for the activation process using token policy, token batch, and token group.

Token Batch and Group

The administrator is required to configure the token batch and token group prior to importing tokens. This can be pre-configured in the Admin Console or created in Token Import Utility (TIU) during token import.

The **Token Batch** is a physical grouping of tokens that are used to determine the batch of tokens with which a particular token policy will be implemented. Refer to [Token Batch](#) for additional details.

The **Token Group**, on the other hand, provides a logical grouping of token devices for ease of management. The token group is used for token assignment, token management, and access control.

- **Token Assignment**

Token group allows system token assignment to assign token in a certain token group to a user.

- **Access Control**

Token group is used to determine which admin can access which tokens.

- **Token Management**

Token group can be used to represent bank branch (e.g. if the tokens are delivered to branch A, then tokens should be moved to token group A). It can also be used in a corporate banking environment, where a group of tokens are given to a company where the company admin can self-manage the token assignment of these tokens (e.g. assign some tokens to token group A that is representing company A).

Refer to [Token Group](#) for additional details.

Token Import Utility (TIU)

Token Import Utility (TIU) is a web-based token management application used for bulk import of tokens into the database via the security server. Refer to [Token Import Utility](#) for additional details.

Token Policy

Each form of token is governed by the relevant token policy as described below:

Token Policy	Description
China OTP Token Policy	This policy is used to manage the behavior of the China standard OTP.
COVR PKI Token Policy	This policy manages the behavior of the COVR PKI token that supports PKI (Public Key Infrastructure). You must register AccessMatrix as a plugin or client application in the COVR Admin Panel before creating COVR PKI token policy. Once COVR token policy is created, an asymmetric public key using ED25519 EC is generated. You can use this public key to register in the COVR Admin Panel for reauthentication.
FIDO2 Token Policy	This policy is used to configure the OIDC Relying Party whitelist for FIDO2 token registration. Multiple Relying Parties can be specified to be whitelisted, or the setting can be left blank to allow all Relying Parties.
HSM (SafeNet ProtectServer) OATH Token Policy	This policy is used for V-Key and other standard OATH tokens such as i-Sprint's OATH token. Note: If this policy is used, then OATH token generation and encryption will be done in the HSM.
HSM (Utimaco CryptoServer) OATH Token Policy	This policy is used for V-Key and other standard OATH tokens such as i-Sprint's OATH token. Note: If this policy is used, then OATH token generation and encryption will be done in the HSM.
OATH Token Policy	This policy governs the behavior of the OATH-based token used for authentication. By default, a predefined OATH token policy called "DefaultOATHTokenPolicy" will be created upon successful installation of AccessMatrix.
Remote Token Policy	This policy contains the endpoint URL information of the remote AccessMatrix server for client authentication purposes.
SHDLDC (Shanghai Donglian Dynamic Code Token Policy	This policy governs the behavior of the Shanghai DongLian K8 token used for authentication.
SMS OTP Token Policy	This policy governs the behavior of the SMS-OTP used for authentication, e.g. OTP length, OTP validity period, and so on. This policy is to be used during login module administration. By default, a predefined SMS-OTP policy called

Token Policy	Description
	"DefaultSmsOtpPolicy" will be created upon successful installation of AccessMatrix.
UAF Token Policy (Deprecated)	This policy governs the behavior of the FIDO UAF token (deprecated) and is used to configure relying party related settings. By default, a predefined UAF token policy called "DefaultUAFTokenPolicy" will be created upon successful installation of AccessMatrix.
VASCO Kernel Parameters (Thales SEE Enabled)	This policy is used to manage the behavior of the Vacman Controller that has been enabled for integration with a Thales HSM SEE (Secure Execution Engine).
VASCO Kernel Parameters	This policy is used to manage the behavior of the Vacman Controller, the core authentication component used in all VASCO server products. By default, predefined VASCO kernel parameters called "DefaultVascoKernelParameters" will be created upon successful installation of AccessMatrix and the required Vasco libraries. Refer to UAS OneSpan Token Implementation Guide for more information.
VASCO Kernel Parameters (SafeNet HSM FM Enabled)	This policy is used to manage the behavior of the Vacman Controller that has been enabled for integration with a SafeNet HSM FM. By default, predefined VASCO kernel parameters (SafeNet HSM FM enabled) called "VascoKernelParameterEracomHSM" will be created upon successful installation of AccessMatrix.
VKey PKI Token Policy	This policy is used to manage the behavior of VKey PKI, a type of mobile token from VKEY that supports PKI (Public Key Infrastructure).

Navigate to **Administration Dashboard > Security Policy > Token > Token Policy**. From this page, the following actions can be performed:

- **Create** a token policy - Click the **Create** button and choose the token policy implementation. Click the **Next** button to display the **Create Token Policy** page of the selected policy. Provide the necessary information and save the changes.
- **Search** for a token policy - Enter a token policy Id and/or token policy name as search criteria and click the **Search** button.
- **View** a token policy - Click on a token policy Id under **Search Token Policy Result** to display the **View Token Policy** page.
- **Edit** a token policy - Click the **Edit** button from the **View Token Policy** page and modify the attribute(s) to change. Save the changes.

-
- **Delete** a token policy - Click the **Delete** button from the **View Token Policy** page. Click **OK** on the confirmation pop-up to delete the token policy.

Token Batch

Navigate to **Operation Dashboard > Token > Token Batch**. From this page, the following actions can be performed:

- **Create** a token batch - Click the **Create** button to display the **Create Token Batch** page. Provide the necessary information and save the changes.



Note: An error message will be displayed if the number of created token batches exceeds the **Max Token Batches** value specified in the server license.

- **Search** for a token batch - Enter a token batch Id and/or token batch name as search criteria and click the **Search** button.
- **View** a token batch - Click on a token batch Id under **Search Token Batch Result** to display the **View Token Batch** page.
- **Edit** a token batch - Click the **Edit** button from the **View Token Batch** page and modify the attribute(s) to change. Save the changes.
- **Delete** a token batch - Click the **Delete** button from the **View Token Batch** page. Click **OK** on the confirmation pop-up to delete the token batch.

View Token Batch Page Attribute Descriptions

Attribute Name	Description
*Token Batch Id	Identification assigned to the token batch.
*Token Batch Name	The name of the token batch.
Description	Brief description for the token batch.
Attributes Tab	
*Token Policy	The token policy assigned to the token batch.
Static Vector	The static vector of the token. Note: This is only used for VASCO Multi-Device.

Attribute Name	Description
Manufacture Time	The date/time when the token is manufactured.
Expiry Time	The token expiry date/time.
Token Attributes Template Id	The identification of the template is used to set token attributes for additional custom information.


Note: Click the **Add Custom Attribute** button to create an additional custom attribute as required.

Token Group

Navigate to **Operation Dashboard > Token > Token Group**. From this page, the following actions can be performed:

- **Create** a token group - Click the **Create** button to display the **Create Token Group** page. Provide the necessary information and save the changes.


Note: An error message will be displayed if the number of created token groups exceeds the **Max Token Groups** value specified in the server license.

- **Search** for a token group - Enter a token group Id, token group name and/or segment as search criteria and click the **Search** button. Tick the **Recursive** checkbox to do a recursive search.
- **View** a token group - Click on a token group Id under **Search Token Group Result** to display the **View Token Group** page.
- **Edit** a token group - Click the **Edit** button from the **View Token Group** page and modify the attribute(s) to change. Save the changes.
- **Delete** a token group - Click the **Delete** button from the **View Token Group** page. Click **OK** on the confirmation pop-up to delete the token group.

System Token Assignment

By default, the system will assign the oldest token first, based on the time each token is placed in a token group. This is particularly useful for hardware tokens that have an expiry date due to the lifespan of their batteries.

For soft tokens, this is not required. Soft tokens have a higher rate of token assignment, as users change their devices (such as mobile phones, etc.) more often than hardware tokens. A soft token vendor such

as V-Key sometimes requires new tokens to be re-provisioned when they update their firmware. To allow higher system-token assignment capacity and to increase performance, it is highly recommended to set:

- **Randomize Reservation Pool** to "true".
- **Database Fetching Order of Reservation Pool** to "None".

These settings fetch the token reservation pool from the database quicker and reduce the chance of conflict and clashing during token assignment for a more efficient system-token assignment process. If the system expects a more extensive concurrent system-token assignment request, increase the value of **Reservation Pool Size**.

View Token Group Page Attribute Descriptions

i Note: ' * ' denotes mandatory attribute.	
Attribute Name	Description
*Token Group Id	Identification assigned to the token group.
*Token Group Name	The name of the token group.
Description	Brief description of the token group.
Version	Version of the token group.
Token Vendor	Vendor of the token group.
Segment	Name of the segment the token group belongs to.
Attributes Tab	
System Token Assignment	
Randomize Reservation Pool	This is to randomize the available tokens when assigning them to a user if there are large system token assignment concurrent requests. Select the setting "true" for soft tokens that require token re-provisioning due to soft token SDK/firmware upgrade. It will fetch the token reservation pool from the database much quicker and reduce the chance of conflict and clashing during the token assignment.
Reservation Pool Size	Set the reservation pool size for random system token assignment. When handling high concurrent requests, set Randomize Reservation Pool to "true" and then set this pool size to at least "20". Default: 10.

Attribute Name	Description
Database Fetching Order of Reservation Pool	This setting specifies the sorting of the fetch from the database to get the reservation pool. Select the setting "None" for soft tokens that require token re-provisioning due to soft token SDK/firmware upgrade. Doing this will greatly reduce the database processing.
Reservation Pool Random Pages	This setting specifies the estimated number of pages of unassigned tokens that will be retrievable from the database. When this attribute is enabled, a random page between the first page and the page of the value specified is selected as the reservation pool. Set to "0" to disable this attribute, which disables pagination such that the top records returned are always returned. Default: 10
Redis Pool Enabled	Use Redis as a centralized pool for unassigned tokens. If this is enabled, the above Reservation Pool settings are ignored and manual assignment is disallowed for this token group. Default: false
Redis Pool Maximum Size	Maximum number of unassigned tokens in the Redis pool. Default: 10000
Redis Pool Minimum Size	Minimum number of unassigned tokens in the Redis pool. When the existing pool size is below the specified value, the producer will fetch tokens from the database and replenish the pool. Default: 100
Redis Pool Expiry in Seconds	If there is no access to the pool for the specified period of time, the pool will be cleared. Default: 600



Note: Click the **Add Custom Attribute** button to create an additional custom attribute as required.

Token Import Utility

Before a token can be assigned to a user, it needs to be imported to the AccessMatrix system first. The Token Import Utility (TIU) allows the administrator to import a token, delete a token, re-assign a token group, and perform various other token-related operations.



Note: TIU will only display the buttons based on the admin rights granted to a user.

To launch the TIU:

1. Click **Cancel** to close the Admin Console login page.
2. Navigate to **Token > Token Import Utility**.

-
3. Enter the first token admin's login Id and password (e.g. tokenAdmin1) and click **OK**.
 4. Enter the second token admin's login Id and password (e.g. tokenAdmin2) and click **OK**. The TIU page will be displayed after a successful login.

Import Token

To import tokens:

1. Click **Import Token**.
2. Select either **Token Seeds Import** or **Activation Password Import**.

i **Note:** By default, the "Token Seeds Import" option is selected. Select Token Seeds Import to perform a normal token import.

3. Specify the VASCO DPX or OATH PSKC file to be used for token import, by completing either of the following sub-steps (a,b):

a. **Upload local file** - Perform the following steps:

- i. Click the corresponding radio button.
- ii. Click **Choose File** and select the file to be uploaded.
- iii. Click **Upload**.

b. **Or specify server file under system temp folder** - Perform the following steps:

- i. Click the corresponding radio button.
- ii. Place the file in your AccessMatrix server's *temp* folder.

i **Note:** This folder is typically located at <ACMATRIX_ROOT>/tomcat/temp/.

- iii. Specify the name of the file (e.g. "example.dpx").

i **Note:** You do not need to specify the path to the temp folder.

4. Set up **Token Batch** by completing either of the following sub-steps (a,b):

a. Select the appropriate token batch from the drop-down list.

i **Note:** Refer to the steps described in [Token Batch](#) on how to create a token batch using the Admin Console.

b. Create a new token batch via TIU, if necessary:

- i. Click **Create** to display the **Create Token Batch** dialog box.
 - ii. Specify the **Token Batch Id**, **Token Batch Name**, and **Token Policy** values.
 - iii. Click **Submit**.
-

5. Set up **Token Group** by completing either of the following sub-steps (a,b):

a. Specify the appropriate token group:

i **Note:** Refer to the steps described in [Token Group](#) on how to create a token group using Admin Console.

- i. Click the drop-down list to display the **Select Token Group** dialog box.
- ii. Click **Search**.
- iii. Select the token group, and click **Add to Selection**.
- iv. Click **Apply**.

b. OR create a new token group via TIU, if necessary:

- i. Click **Create** to display the **Create Token Group** dialog box.
- ii. Specify the ***Token Group Id**, ***Token Group Name**, **Token Vendor**. If necessary, specify the **Description** and **Segment** as well.
- iii. Click **Submit**.

6. Select the default **Token Status** of tokens after import.

i **Note:** The default value is "Activated".

7. [OPTIONAL] Tick the **Issuance Token Group for System Token Assignment** checkbox if you wish to use this feature.

i **Note:** For more information, please refer to [Issuance Token Group](#).

8. Specify the **Transport Key**.

i **Note:** KCV and key count are read-only and auto-generated by the system.

9. Select the action to be taken in the event of a conflict.

10. [OPTIONAL] Enter the job description where the user can describe more about the current importing job.

11. [OPTIONAL] Click **Preview Token** to preview the tokens to be imported. A list of tokens with their details will be displayed.

12. Click **Import Token** to start the token import process.

Issuance Token Group

During Step 2 of the token import process, you may choose to tick the optional **Issuance Token Group for System Token Assignment** checkbox. If enabled, tokens will be assigned to an issuance token

group when they are imported. The issuance token group will be the same token group that was indicated in the **Token Group** field.

An issuance token group is a token group used specifically for system token assignment (issuance) purposes. Instead of assigning individual tokens to a user, the administrator may opt to perform token assignment via issuance token group. The system will assign a token from the specified issuance token group, and the token will be removed from the issuance token group once the assignment is completed.

Using issuance token groups for system token assignment may help to improve performance, especially when large quantities of tokens need to be assigned.

**Notes:**

- The same token group can be used as both the regular token group and the issuance token group of a token. This will be the case for tokens imported via TIU when the **Issuance Token Group for System Token Assignment** option is enabled.
- After a token is assigned by the system to a user via issuance token group, the token is removed from that issuance token group. However, it is not removed from the regular token group.
- Tokens that belong to an issuance token group cannot be manually assigned. They must be assigned using the system token assignment via issuance token group method.
- When searching for tokens that have yet to be assigned, you can filter the search results by the issuance token group.

Import Token PIN

TIU can be used to import token PINs into the AccessMatrix system. Certain models of tokens require associated token PINs to be imported into the AccessMatrix system before the tokens can be assigned and used by the users. Importing of the token PINs into AccessMatrix require the associated tokens to be imported into the AccessMatrix beforehand. For this type of tokens, the following should be prepared:

- Token EPV file that contains the encrypted token PINs information.
- The corresponding EPV Transport Key to decrypt the token PINs in the EPV file.

On the TIU page, click the **Import PIN** button and specify the EPV file and the keys required. Then, click the **Import PIN** button on the **Import PIN** page to start the import process.

Print Token PINs

TIU can be used to print token PIN mailers from the imported tokens. Token PIN mailer is a controlled document on which the initial Token PIN of the user is printed and mailed.

List Token Bulk Jobs

The TIU allows for administrators to view the list of token bulk jobs, and other detailed information for specific bulk jobs.

To view the list of token bulk jobs:

- On the TIU page, click the **List Token Bulk Jobs** button.

This page provides some basic information about the token bulk jobs:

- **Bulk Job Id**
Indicates the bulk job Id that has been generated by the server.
- **Description**
Indicates filename of the uploaded file, as well as the description provided by the user when setting up the import process.
- **Type**
Indicates the bulk job type (e.g. Token Import).
- **Status**
Indicates the status of the bulk job.

Status	Description
New	The bulk job has been scheduled but has yet to start running the process.
Running	The bulk job is currently running.
Completed	The bulk job has been completed.
Terminated	The bulk job was terminated due to token import failure.

- **Server Id**
Indicates the server Id from which this bulk job was created from.
-

- **Creator**

Indicates the user who created the bulk job.

- **Created Time**

Indicates the time for when the bulk job was created.

To view more information for a specific bulk job:

- Click the bulk job Id specified in the **Bulk Job Id** column.

This page displays additional information regarding the selected bulk job:

- **Result**

Indicates the result of the import token process as well as the number and range of tokens imported.

- **Progress Log**

Indicates the progress of the bulk job, and completed stages of the token import process.

- **Start Time** and **End Time**

Indicates the start and end time of the bulk job.

Reassign Token Group

To reassign imported tokens from one token group to another:

1. On the TIU page, click the **Reassign Token Group** button.
2. Select the **Token Vendor** from the drop-down list.
3. Select the token batch from the **Token Batch** drop-down list and click on the **Retrieve serial numbers** button to locate the first and last token serial number in the selected batch. The **Token Serial Number From** and **Token Serial Number To** fields will be supplied accordingly once the system is able to locate the serial numbers.
4. The range of token serial numbers for the **Token Serial Number From** and **Token Serial Number To** fields can also be manually entered.



Note: If both the starting and ending serial numbers are not specified, this will imply that all the existing tokens belonging to the source token group are to be reassigned to the target/destination token group.

5. Select the desired token groups from and to which the tokens are to be reassigned in the **Token Group Reassign From** and **Token Group Reassign To** fields respectively.
-

-
6. To reassign the tokens to a new group, the administrator may create a new token group by clicking the **Create** button where a **Create Token Group** dialog box is displayed. Specify the necessary information and click **Submit** to create the new token group.
 7. Specify the token status of the tokens to be reassigned upon successful reassignment in the **Token Status (after assignment)** field.
 8. [OPTIONAL] Click on the **Preview Token** button to preview the list of matched tokens based on the selected **Token Group**, **Token Batch**, **Token Vendor** and the range of the **Token Serial Number**.
 9. Click the **Reassign** button to start the token group reassignment operation.
 10. Upon successful token group reassignment, the system will display the list of tokens successfully reassigned to the target/destination token group.

Delete Token Bulk

On the **TIU** page, click the **Delete Token** button. Select the token vendor of the tokens to be deleted from the **Token Vendor** drop-down list and specify the range of token serial numbers to delete in the **Token Serial Number From** and **Token Serial Number To** fields respectively. The administrator may tick the **Unassign Token** option so that if the token to be deleted is currently assigned to a user, the system will unassign the token from the user first and then only proceed to delete the token. Otherwise, the system will fail to delete the token since it is currently assigned to a user. Click the **Delete** button to start the token deletion process.

Thales Vasco Key Import

Refer to the [UAS OneSpan SEE with nShield Connect Integration Guide](#) for the import steps.

View Imported Token

Navigate to **Operation Dashboard > Token > Token**. This page contains the list of imported tokens to the system and the following actions can be performed:

- **Search** for a token - Enter the token serial number, token status, token vendor, token batch Id, issuance token group Id, and/or token group Id as the search criteria and click the **Search** button.
-

- **Advanced Search** for a token - Click the **Advanced Search** button and enter the following criteria: token serial number from, token serial number to, serial number begins with, token group, issuance token group Id, status, and/or records/1 pages and click the **Search** button.
- **View** a token - Click on a token serial number under **Search Token Result** to display the **View Token** page where token details will be displayed. This page displays buttons that can perform actions such as resynchronization of tokens, token status change, etc.

View Token Page Attribute Descriptions

i	Note: '*' denotes mandatory attribute
Attribute Name	Description
Token Serial Number	Identification number assigned to the token.
Version	The token revision.
Issuance Token Group for System Token Assignment	The issuance token group that the token belongs to. Clicking the value will display the issuance token group's details.
Token Group	The token group of the token. Clicking the value will display the token group's details.
*Token Model	The token model of the token.
Token Batch	The token batch of the token. Clicking the value will display the token batch's details.
Token Vendor	The vendor of the token.
Assigned To	The user Id whom the token is assigned. Clicking the value will display the user's details.
Token Status	Status of the token.
Attributes Tab	
There is no available attributes, however, custom attribute(s) may be added.	
Status Change History Tab	
This tab displays the status change history of the token.	
i	Note: Additional tab(s) showing the token's details (such as Application Id, PIN Length, Response Type , etc.) will be displayed depending on the token type.

Global Setting - Token Management

The token management attributes allow the administrator to specify the features available for Token Import Utility (TIU) whether to allow the token to be verified after the token has been resynchronized, whether to allow for sharing of a token among the users, the mapping of Token Vendor-Model-Function, etc.

Mapping of Token Vendor-Model-Function

The mapping of Token Vendor-Model-Function determines the functions supported by a particular type of token model provided by a particular vendor. It also allows the administrator to define new token models together with the supported functions. Several among the token-related functions supported include unlock token, reset password, resynchronize token, etc.

To access this page, navigate to **Administration Dashboard > Configuration > Global Setting** and click the **Token Management** tab.

Token Management Configuration Parameters

Attribute	Type	Value	Description
Token Import - Number of key components	int	1 to 3	Number of key fields to be entered when importing token DPX file. Default: 1
Token Import - Show extended keys	boolean	True or False	Displays the Operator Key and Administrator Key fields during the token import if this attribute is enabled. Default: false
Verify Token After Resync	boolean	True or False	Token verification is required after resync if the attribute is set to "true". Default: false
Token Vendor Model Function Map	complex		It maps token functions to a list of token models (each model can be a regular expression mask) which support the function.

Attribute	Type	Value	Description
			Refer to the table below for the configuration of Token Vendor Model Function Map .
Preload All Token Batches	boolean	True or False	Preload all token batches for faster access. Default: false
Token Status Display List	multi-selection	All, Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The status set here will be the status displayed in the Token Inventory report if the Token Status Display List attribute in the Token Inventory Report page is set to "Filter by token status display list in global setting". Default: Unassigned
Token Status After Unassign	list	[No Change], Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The token status that will be set after token unassignment. Default: Unassigned
Token Status After Assignment	list	[No Change], Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The token status that will be set after token assignment. Default: [No Change]
Token Status (if threshold exceeded)	list	[No Change], Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The status that will be set if token verification failed count reaches up to the maximum allowed number of tries. Default: Locked
Token Status for usable functions like verify OTP/signature/response etc.	list	Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The statuses for a token to be considered in an operational or usable state. Default: Activated & PreActivation

Attribute	Type	Value	Description
Token Assignment Restriction	list	No Restriction, Restrict to One Token Assignment Per User, Restrict to One Token Assignment Per User Per Vendor, Restrict to One Activated Token Assignment Per User Per Vendor	• "No Restriction" - Has no token assignment restriction. • "Restrict to One Token Assignment Per User" - Allows you to assign a token for each user. If another token is assigned to the user, the server will prompt "Current policy does not allow more than one token assigned per user". • "Restrict to One Token Assignment Per User Per Vendor" - Restricts each user to only 1 token assignment from each vendor. If there are multiple tokens from the same vendor that are assigned to a specific user, the server will prompt "Current policy does not allow more than one token assigned per user per vendor". • "Restrict to One Activated Token Assignment Per User Per Vendor" - Restricts each user to only 1 activated token assignment from each vendor. If there are multiple activated tokens from the same vendor that are assigned to a specific user, the server will prompt "Current policy does not allow more than one activated token

Attribute	Type	Value	Description
			assigned per user per vendor".
Token Status for tokens to be assignable	list	Activated, Battery Expired, Damaged, Deactivated, Deleted, Elevated, Expired, Faulty, Frozen, In_Transit, Invalidated, Locked, Lost, PreActivation, Returned, Suspended, Unassigned, or Unuspended	The statuses for a token to be assignable. Default: Activated, PreActivation & Locked
DPMobileGateway Service URL	string	http://127.0.0.1:8080/DP4MobileGateway/DPMobileService	Specifies the HTTP URL for DPMobileGateway Service. It will be used during push notification sending request for DPMobile. Format: http://<ip_address>:<port>/context/service
Number of Assignable Tokens to User	int	0-10	The number of assignable tokens that can be assigned to a user. If the value is set to "0", then unlimited tokens can be assigned to a user. Default: 5

Token Vendor Model Function Map

Token Vendor	Attribute	Type	Value
VASCO	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & "."
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
	CHANGE_ALLOW_ACTIVATION	string	Default: "MOB[0-9]{2,5}"

Token Vendor	Attribute	Type	Value
VASCO_FM_ERACOM	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & " *"
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
	CHANGE_ALLOW_ACTIVATION	string	Default: "MOB[0-9]{2,5}"
VASCO_FM_SAFENET	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & " *"
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
	CHANGE_ALLOW_ACTIVATION	string	Default: "MOB[0-9]{2,5}"
VASCO_FM_THALES	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & " *"
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: (DP)?GO[0-9]+
	CHANGE_ALLOW_ACTIVATION	string	Default: "MOB[0-9]{2,5}"
VASCO_MULTI_DEVICE	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & " *"
	UNLOCK	string	Default: "DP[0-9]{3,5}"

Token Vendor	Attribute	Type	Value
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
VASCO_MULTI_DEVICE_FM_SAFENET	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & ".*"
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
VASCO_MULTI_DEVICE_FM_THALES	RESYNC	string	Default: "(DP)?GO[0-9]+", "DP[0-9]{3,5}" & ".*"
	UNLOCK	string	Default: "DP[0-9]{3,5}"
	RESET_STATIC_PWD	string	Default: "(DP)?GO[0-9]+"
OATH	RESYNC	string	".*"
HSMOATH	RESYNC	string	".*"
SHDLDC	RESYNC	string	".*"
	UNLOCK	string	".*"
	ACTIVATE	string	".*"

Token Inventory Monitoring

Token Inventory Monitoring runs daily at 2 A.M. to monitor assignable tokens. A notification message is sent when the assignable tokens are almost depleted based on the specified threshold value. A default event notification policy, "DefaultTokenInventoryNotificationPolicy", and a default message template, "DefaultTokenInventoryTemplate", are already pre-created.

To access this page, navigate to **Administration Dashboard > Configuration > Token > Token Inventory Monitoring**. From this page, the following actions can be performed:

- **Create** a token inventory monitoring - Click the **Create** button to display the **Create Token Inventory Monitoring** page. Provide the necessary information and save the changes.
- **Search** for token inventory monitoring - Enter a Token Inventory Monitoring Id and/or Token Inventory Monitoring Name as search criteria and click the **Search** button.
- **View** a token inventory monitoring - Click on a token inventory monitoring Id under **Search Token Inventory Monitoring Result** to display the **View Token Inventory Monitoring** page.
- **Edit** a token inventory monitoring - Click the **Edit** button from the **View Token Inventory Monitoring** page and modify the attribute(s) to change. Save the changes.
- **Delete** a token inventory monitoring - Click the **Delete** button from the **View Token Inventory Monitoring** page. Click **OK** on the confirmation pop-up to delete the token inventory monitoring.

Create Token Inventory Monitoring Page Attribute Descriptions

	Note: '*' denotes mandatory attribute
Attribute Name	Description
*Token Inventory Monitoring Id	Identification assigned to the token inventory monitoring. (uneditable)
*Token Inventory Monitoring Name	The name of the token inventory monitoring.
Description	Brief description of the token inventory monitoring.
Version	The token inventory monitoring revision. (uneditable)
Attributes Tab	
Token Group(s) To Monitor	Refers to the token group(s) to monitor.
Threshold	Sets the token count threshold. The assignable tokens that are not assigned to any users will be counted. (Assignable tokens are those with status defined in Administration Dashboard > Configuration > Global Setting > Token Management > Token Status for tokens to be assignable). Event notification with a "Failed" result will be fired when the number of tokens counted is less than the threshold. Event notification with a "Success" result will be fired when the number of tokens counted is more or equal

Attribute Name	Description
	to the threshold. Default: 1000
Event Notification Policy	Specifies the event notification policy to use for the alert message. Note: Only event notification policies with the Event Category "Token" and the Event type "InventoryMonitoring" are selectable. Default: DefaultTokenInventoryNotificationPolicy

Token Import Utility (TIU) Configuration

TIU Configuration allows the administrator to define attributes such as maximum tokens allowed for each token group reassignment and the list of token groups available for selection during the token import process.

To access this page, navigate to **Administration Dashboard > Configuration > Token > TIU**

Configuration. From this page, this action can be performed:

- **Edit** the TIU configuration - Click the **Edit** button from the **View Configuration** page and modify the attribute(s) to change in **Edit Configuration** page. Save the changes.

Edit Configuration Page Attribute Descriptions

 Note: '*' denotes a mandatory attribute	
Attribute Name	Description
*Configuration Id	Identification assigned to the TIU configuration. (uneditable)
*Configuration Name	The name of the TIU configuration.
Description	Brief description for the TIU configuration.
Version	The TIU configuration revision. (uneditable)
Implementation	Implementation type. (uneditable)
Attributes Tab	
Max Tokens For Token Group Reassignment	Controls the maximum number of tokens for token group reassignment in TIU. Value should not be less than 1. Default: 1000

Attribute Name	Description
Allow overwriting existing tokens option	Setting to "true" will enable Overwrite existing tokens during token import activity option on the Import Token page of TIU. Default: false
Log imported token status to CSV file	All CSV files will be logged in the folder <ACMATRIX_ROOT>\tomcat\work\bulkjobs\logs\ Default: false
Token Status Display List	Only the selected values will be displayed in the Token Status attribute of the Reassign Token Group page. This setting will overwrite the same attribute in the Global Setting > Token Management tab. Note: If "-" is selected, it will follow the list specified in Global Setting.
List these token groups only Tab	
Only the specified token group(s) will be listed under the Token Group list in TIU.	

3.2. Token Administration

The administrator can carry out important tasks associated with tokens such as token assignment/unassignment to/from the user, token delivery to users, changing of token status, token sharing, etc.

Token Assignment

The token can be assigned and unassigned through Admin Console. It can either be through system assignment or by manually entering the token serial number.

Assign Tokens

1. Navigate to **Operation Dashboard > User & Segment > User** and search for the user to assign with a token. Create a new user, if necessary.
2. Click **Token Assignment** and select a **Token Reassignment** option:
 - To allow the system to assign an available token from the selected **Token Group**, select the **System Assigned** option and select the group from the **Token Group** drop-down list.
 - To manually assign a token to a user, select the **Enter Serial Number/ID** option and enter the token serial number. Select the vendor from the **Token Vendor** drop-down list.
3. Click **OK** for the token assignment to be performed. An information message for a successful token assignment will be displayed. Click **Yes** to continue with token assignment; otherwise, click the **No** button.
4. The assigned token will be displayed under **User > Assigned Token** tab.

Unassign Tokens



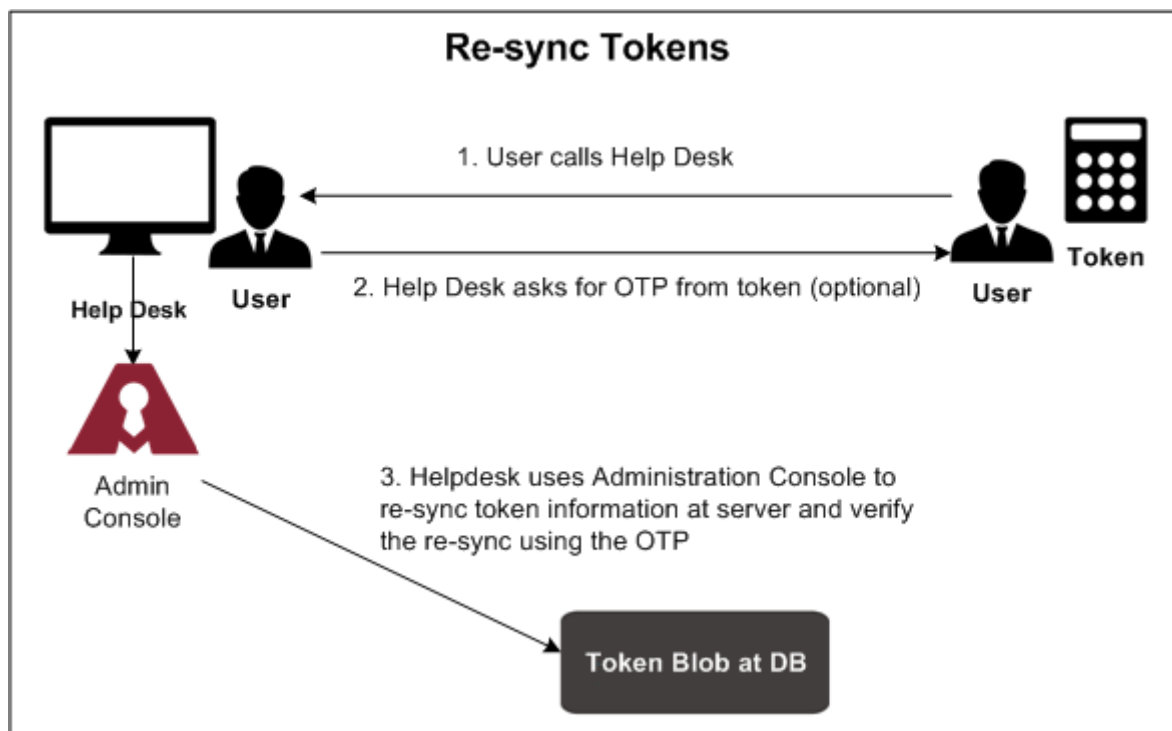
Note: If a user needs to be reassigned with another token, then the token previously assigned to the user must be unassigned first.

1. Navigate to **Operation Dashboard > User & Segment > User** and search for the user whose token will be unassigned.
 2. Click the **Token Assignment** button and select **No Assignment** from **Token Reassignment** options.
 3. Tick the checkbox beside the token serial number to unassign.
-

-
- Click **OK** for the token unassignment to be performed. An information message for successful token unassignment will be displayed. Click **Yes** to continue with a token assignment; otherwise, click the **No** button.

Resync Token

Token may be out-of-sync with the security server due to the time drift from the GMT time. To resolve this problem, the user can request the HelpDesk to resynchronize the token.



To resynchronize the token:

- Navigate to **Operation Dashboard > Token > Token** at the left pane.
 - Specify the search criteria and search for the token to resync.
 - Click the token serial number to resync under the **Token Serial Number** column.
 - Click **Resynchronize** button. The **Resynchronize Token** page is displayed.
 - Select the **Application Id** from the drop-down list and select the **New Status** of the token after a resync.
 - Click the **Resynchronize** button to start the resynchronization process.
-

Change Token Status

To change the token status:

1. Navigate to **Operation Dashboard > User & Segment > Token**.
2. Specify the search criteria and search for the token to change.
3. Click the token serial number under the **Token Serial Number** column.
4. Click **Change Status** from the **View Token** page.
5. Select the token status from the **Token Status** drop-down list. (Refer to the table below for the list of token states.)
6. Optionally, enter the **Remark** for the change of token status.
7. Click **Change Status** to start the process.

Token States

The token state reflects the current status of the physical token. Each token status can be changed depending on certain conditions or events.

List of token states:

Token Status	Description
Activated	The status when the token is active and can be assigned/unassigned to/from the user. This is the default status of a token ready for assignment and the status when the token is usable for authentication.
Battery Expired	The status when the token's battery is already expired. This is for hardware token where the battery life is defined in the token batch.
Damaged	The status when the token has been reported as damaged.
Deactivated	The status when the token instance is lost. Used for VASCO Multi-Device tokens. Note: Any token instance that has been flagged as deactivated will not be usable for any token related operations.
Deleted	The status when the token has been deleted.
Elevated	Currently, it's unused unless defined by the client application.

Token Status	Description
Expired	The status when the token has expired and cannot be used for authentication. This is used for SMS OTP where OTP is expired when used after the defined expiry time in token policy.
Faulty	The status when the token is faulty and cannot be used anymore.
Frozen	This temporarily disables the token authentication. Note: Token is frozen when it is temporarily suspended. The user will not be able to use it until it goes back to the assigned state.
In_Transit	This is an intermediate state where the token is currently being transited.
Invalidated	The status when the token has been invalidated and cannot be used anymore.
Locked	The status when the token is locked and cannot be used for authentication. Note: Token status will be set to "Locked" if the token policy attribute Lock Token If Bad Verification Count Is More Than has been fulfilled.
Lost	The status when the token has been reported as lost.
PreActivation	The status when the token needs to be activated for assignment to a user.
Returned	The status when the token has been returned to the token issuer.
Suspended	This temporarily suspends the token authentication.
Unassigned	The status when the token was assigned to a user before and the user does not require the token anymore. Currently, it has been unassigned from the user, but the user has not returned the token. In this case, the token status should stay in an unassigned state until the token is returned. A returned token can then be marked as active and can be assigned again to another user.
Unsuspending	The status when the token has been changed from the "Suspended" state to "Unsuspending" state.

Reset Token Static Password

Static password refers to the password remembered by the user and verified at the security server. It is different from the token pin where it is verified by the token. In a certain scenario, E2EE password will be used as the static password. Alternatively, VASCO token for example also offers static password verification using the token blob. A static password is usually set during token initialization.

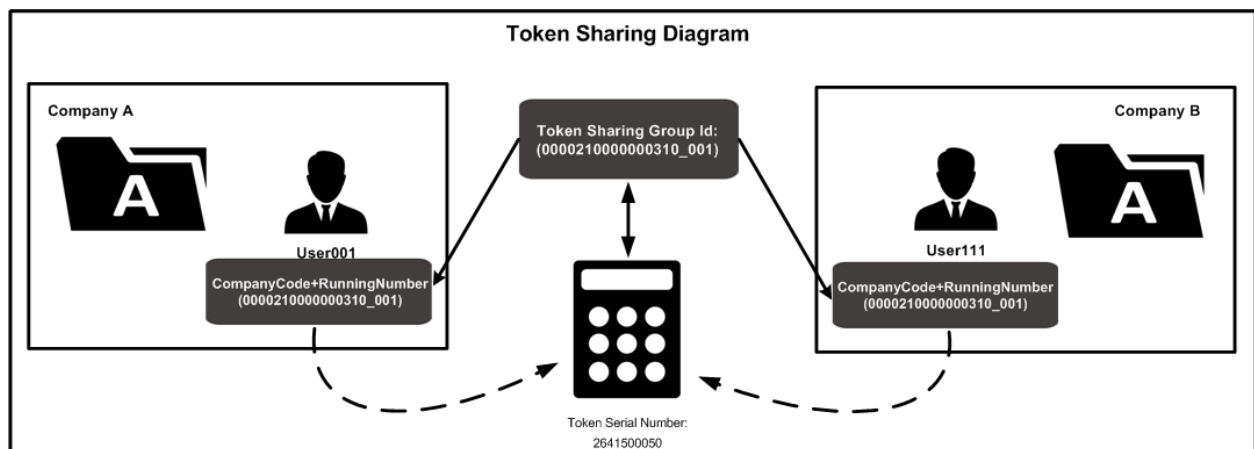
To reset the token password:

1. Navigate to **Operation Dashboard > Token > Token** and search for the token to reset.

2. Select the token serial number and the **View Token** page will be displayed.
3. Click **Reset Password** and enter values to **Enter new static PIN** and **Re-enter new static PIN** fields.
4. Click **Reset Password** button to start the reset token password process.

Token Sharing

Token sharing is a feature that allows a token to be shared by multiple users. All users must have common information which should not change over time called **Token Sharing Group ID**.



Note: Prior to token sharing, it is assumed that tokens are already imported. Refer to [Import Token](#) section for the steps.

To configure token sharing:

1. Create a new user or edit an existing user, e.g. "User001".
2. Specify the **Token Sharing Group ID** value, e.g. "0000210000000310_001". Save the changes.

Note: This attribute must be set first before assigning a token to a user.

3. On the **View User** page, click **Token Assignment** button.
4. Assign a token to a user. Choose either system assignment or by entering the token serial number/Id. Refer to the [Token Assignment](#) section for details.
5. If the token assignment is successful, the token will be listed under the **Assigned Token** table. The table will display the following values: **Token Serial Number**, **Token Vendor**, **Token Model**, and **Token Sharing Group ID / Members**. Under the **Token Sharing Group ID / Members** column, the group Id will be displayed with the list of users sharing the token, e.g. "0000210000000310_001 - User001".

Note: The token is in sharing mode already.

6. To share with another user (e.g. User111), set the same **Token Sharing Group ID** for this user. The token previously assigned will be automatically shared with the second user (User111). The **Token Sharing Group ID / Members** column will display "0000210000000310_001 - User001 - User111".

Unassign and Unshare Tokens

Unassign Token

To unassign token, perform the steps described in [Token Assignment - Unassign Tokens](#) section. If the unassignment is successful, the token will be unassigned from all users with the same Token Sharing Group ID. Do note that assigning a new or the same token back to any one of the users will automatically assign the token to all users with the same Token Group Sharing ID. This is because the token is not directly mapped to the user Id but is mapped with Token Group Sharing ID.

Unshare Token

Unassignment only unassign the token from the user, but it does not remove the user from token sharing. To completely remove the token sharing from the user, ensure that the token is unassigned first, then the Token Sharing Group ID value is removed from the user.

Note: Unsharing the token without the unassignment is not recommended as the token will still have the relationship with the Token Sharing Group ID.

Multiple Token Sharing Groups

AccessMatrix UAS users use their Token Sharing Group IDs (TSG IDs) for token sharings can have multiple Token Sharing Group IDs for different types of tokens, whether they are soft tokens or hard tokens. Therefore, a user can have one or more tokens in one Token Sharing Group (TSG) or multiple TSGs.

The administrator can assign the tokens to the individual users (i.e. User Level) or the Token Sharing Groups (TSG Level). In some cases, the administrator assigns tokens to individual users (i.e. User Level) when no sharing is required. However, when multiple token sharing groups are needed, the administrator will assign the tokens to the Token Sharing Group IDs (TSG Level).

Assign Multiple Token Sharing Group IDs

You have already created new users and assigned tokens to the respective users. If not, refer to [Token Assignment - Assign Tokens](#) for the steps.

Assign Token to Token Sharing Group (TSG) Level

1. Navigate to **Operation Dashboard > User & Segment > User** and search for the user to add with a Token Sharing Group ID (TSG ID).
2. Click **Edit**, type the TSG ID in the field beside the **Add** button, and click **Add**. You can add multiple TSG IDs to the list for a user. Click **Save**.
3. Click the **Token Assignment** button at the bottom of the page and select a **Token Reassignment** option:
 - To allow the system to assign an available token from the selected **Token Group**, select the **System Assigned** option and select the group from the **Token Group** drop-down list.
 - To manually assign a token to a user, select the **Enter Serial Number/ID** option and enter the token serial number. Next, select the vendor from the **Token Vendor** drop-down list.
4. Next, click **Select Token Sharing Group Id** and select a TSG ID from its drop-down list.
5. Click **OK**. The assigned token will be displayed under the **Assigned Token** section.

Assign Token to Individual User Level

1. Navigate to **Operation Dashboard > User & Segment > User** and search for the user to assign a token to at the user level.
 2. Click **Edit**, type the TSG ID in the field beside the **Add** button, and click **Add**. You can add multiple TSG IDs to the list for a user. Click **Save**.
 3. Click the **Token Assignment** button and select a **Token Reassignment** option:
 - To allow the system to assign an available token from the selected **Token Group**, select the **System Assigned** option and select the group from the **Token Group** drop-down list.
-

-
- To manually assign a token to a user, select the **Enter Serial Number/ID** option and enter the token serial number. Next, select the vendor from the **Token Vendor** drop-down list.
4. Next, select **No Sharing** at the **Token Sharing** section.
 5. Click **OK**. The assigned token will be displayed under the **Assigned Token** section.

Multiple Token Sharing Groups for Remote Tokens

Multiple token sharing groups are supported for the remote tokens. You can assign, unassign, and re-assign multiple token sharing groups for the VIRTUAL_REMOTE_TOKEN vendors. Tokens are imported into the remote AccessMatrix server while the application is connected to the local AccessMatrix server.

Token assignments of the remote tokens are set up on two AccessMatrix servers (Local and Remote servers). At the local server, tokens are assigned to the TSG IDs; tokens are assigned to the dummy TSG IDs at the remote server.

Preparations for the two servers (local and remote servers)

1. Set up a local AccessMatrix server:
 - Copy */sdk/remotetoken/am-remote-token-core.jar* to *tomcat/webapps/am5/WEB-INF/lib/*.
 - Copy */sdk/remotetoken/am-remote-token-plugin.jar* to *tomcat/plugins/*.
 - Copy */sdk/remotetoken/field-definitions-remotetoken.xml* to *tomcat/webapps/am5/WEB-INF/classes/*.
 - Copy and replace */sdk/remotetoken/wpecustom-remotetoken.properties* to *tomcat/webapps/am5/WEB-INF/classes/wpecustom.properties*.
 - Edit *tomcat/webapps/am5/WEB-INF/classes/amsystem.properties* to add the following:
am.field_definition.file.0=field-definitions-remotetoken.xml.
 - Edit *tomcat/webapps/am5/WEB-INF/web.xml* and uncomment
com.isprint.am.remotetoken.TinyRemoteTokenService.
 2. Start both the local and remote servers.
 3. Log in to the local server, then [Set Up Remote Token Policy in Admin Console](#), specify **Remote Endpoint URL** as *http://<HOST>:<PORT>/am5/rest*.
 4. [Create a token batch](#), i.e. "RTB", and then select the previously created token policy from the drop-down list.
-

Token Assignment Record Data Integrity Checking Policy

The data integrity checking policy governs the behavior of AccessMatrix's data integrity checking mechanism. This mechanism verifies the data integrity of AccessMatrix users' token assignment records to ensure they have not been tampered with.

By default, a system-defined data integrity checking policy is created upon the successful installation of AccessMatrix. The administrator is allowed to view and configure the data integrity checking policy settings. To locate the **View Policy** page of this policy, navigate to **Administration Dashboard > Security Policy > Admin & Audit > Data Integrity Checking Policy** on the left pane. The policy configurations can be edited by clicking the **Edit** button on this **View Policy** page.

View Data Integrity Checking Policy Page Attribute Description

Attribute Name	Description
*Policy Id	System pre-defined data integrity checking policy Id.
*Policy Name	Name of data integrity checking policy.
Description	Description of data integrity checking policy.
Version	Data integrity checking policy entry version.
Implementation	Implementation type of the data integrity checking policy.
Attributes Tab	
Generate data integrity signature for Token Assignment	When this attribute is enabled, AccessMatrix generates a data integrity signature whenever a token is assigned to or unassigned from a user. This signature is used to verify the data integrity of the user token assignment records. Default: false
Data Integrity Key	The key used to sign the token assignment audit data fields for the purpose of data integrity checking.

To enable data integrity checking:

1. Set **Generate data integrity signature for Token Assignment** to "true".
 2. Select an appropriate data integrity key for **Data Integrity Key**.
 3. **Save** the changes.
-

Once the above policy is configured, the REST API `/rest/token/data-integrity/check/assignment` can be called to carry out the data integrity checking of a user's token assignment records. A successful response indicating that the user's token assignment records are untampered contains:

- The data integrity signature.
- A "true" value for the "verified" field.

For example:

Request Example	
JSON	<pre>{ "sessionToken": "<Admin session token>", "userId": "<user Id>" }</pre>
Note: • sessionToken: The session token of an admin user, also called an admin session. An admin session is one of various authentication methods for admins/application servers to prove that they have the necessary authorization to use certain AccessMatrix REST APIs. Other methods include: ○ Dynamic agent session ○ Access token Refer to AccessMatrix REST API Authentication and Authorization Guide - AccessMatrix REST API Authentication or more information on each of these methods.	
Response Example	
JSON	<pre>{ "result": { "signature": "SHA2,M7ajKb5JaGwg5yX0/ckX8g8iZXLrkVTV0TEJUsIvNM=", "verified": true }, "amProcessingTimeMillis": 71 }</pre>

Refer to the [AccessMatrix REST API Specification](#) for the full API details.

3.3. Report

The following sections provide details and instructions on using the various UAS-related reports found under **Report Dashboard** in the Admin Console:

- [System & Admin](#)
- [Token & SMS](#)

3.3.1. System & Admin

System & Admin Reports include System configuration & Administration related information.



Note:

- Reports can be generated while leaving a field blank or entering a ' * ' wildcard, which will search for all available results.
- Attributes marked with ' * ' denote mandatory; a value must be entered to generate the report.

Real Time Reports

Real-time reports allow the administrator to perform live monitoring of the system and user events.

Below are the system and admin related real-time reports.

Token Assignment Count By Token Vendor Report

In UAS, token usage billing is based on the total tokens that have been assigned to users for the specific token vendor before a specific time. This report is used to show exact token assignments for a period of time.

To successfully generate this report, the license with enabled token usage billing feature and the keystore to sign the report must be distributed first to the customer. The administrator imports this license and configures the reported signature in the Admin Console. The report signature configuration is used to sign the generated report to assure data integrity.

Perform the following steps to enable and configure settings related to this report:

1. Navigate to **Administration Dashboard > System > Server & Cache > Server** and select the server Id from the **Search Server Result**. Select the **Licensed Features** tab, click **Edit** and import the AccessMatrix server license. Ensure that the **Enable Token Usage Billing** license feature is set to "Yes".
 2. Click **Save** to apply the license. Log out and log in again to Admin Console for the license to take effect.
 3. Navigate to **Administration Dashboard > Configuration > Report**, click **Edit**.
 4. Configure the **Report Signature Configuration** attributes:
-



Note: The **Report Signature Configuration** section will only be displayed if the **Enable Token Usage Billing** license is enabled.

- a. Upload the keystore **.jks** file.
 - b. Enter the keystore's corresponding password and alias.
5. Click **Save**.

To navigate to the **Token Assignment Count By Token Vendor Report** page, select **Report Dashboard > System & Admin > Real Time** and click **Token Assignment Count By Token Vendor** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects the reported date, token vendor, token model, and the number of tokens assigned per token vendor.



Note: Currently, this report can only be generated as a CSV file. Refer to [Token Assignment Count By Vendor Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute Name	Description
Reported Date	The date the report is generated.
Token Vendor	The vendor of the token.
Token Model	The token model of the token.
Token Assigned Count	Indicates the number of tokens assigned per token vendor for a period of time.

3.3.2. Token & SMS

Token & SMS Reports include Token activities & SMS OTP related information.

**Note:**

- Most reports can be generated while leaving a field blank or entering a ' * ' wildcard, which will search for all available results, except for certain scenarios using the [Token Inventory Report By Summary](#).
- Attributes marked with ' * ' denote mandatory; a value must be entered to generate the report.

Real Time Reports

Real-time reports allow the administrator to perform live monitoring of the tokens and SMS events.

Below are the token and SMS related real-time reports.

Token Inventory Report

This report provides a summary of tokens belonging to a token vendor and specific token model. It provides the list of token status and the number of tokens assigned and unassigned per status. There are several report types that are available, which allows for filtering by token batch, token group, or token summary. To view the sample reports for each report type, you may click on the following links:

- [By Summary](#)
- [By Batch](#)
- [By Group](#)

To generate the report:

1. Navigate to **Report Dashboard > Token & SMS > Real Time** and click on **Token Inventory** on the left pane.
2. Select a **Report Type** (e.g. "Summary", "By Batch", "By Group") and specify the relevant filtering criteria.



Note: To find out more about the filtering criteria for each **Report Type**, refer to [Report Filtering Criteria](#).

3. Select the file format for the report using the drop-down list (e.g. "HTML", "PDF", "CSV").
 4. Click the **Generate** button.
-

Generally, the contents of the generated report will reflect the token status, token assigned, token unassigned and the total number of tokens:

Attribute	Description
Token Vendor	Indicates the token vendor.
Token Status	Indicates the status of the token.
Token Assigned	Indicates the number of tokens assigned per token vendor.
Token Unassigned	Indicates the number of tokens unassigned per token vendor.
Total Number	Indicates the total number of tokens per token vendor.

Report Filtering Criteria

There are various filters available to each token inventory report type. You may refer to the following table for the detailed description for each criteria.

Criteria	Description
Token Inventory By Summary Report	
This report does not allow wildcards or blank values for the Token Model criteria when Token Group Filter Mode is set to "Include" or "Exclude".	
Token Model	Specifies the token model to be filtered. Note: If Token Group Filter Mode has been specified as "Exclude" or "Include", a specific token model must be entered in this field.
Include Pie Chart	If specified as "Yes", the report includes pie charts of the tokens assigned and unassigned respectively. Note: The report can only be generated in PDF format if this has been enabled.
Token Status Display List *	If specified as "Filter by token status display list in global setting", the report will only include tokens with certain statuses as defined in the Token Status Display List attribute in the Token Management settings. To know more about how to configure this, refer to Global Setting - Token Management .
Token Group Filter Mode	Specifies the filtering rule that will be applied to the token group specified in Token Group Id . Default: None Note: When using the values "Exclude" or "Include", the Token Model will need to be specified.

Criteria	Description
Token Group Id	Specifies the token group to be included in the report. Multiple token groups can be included.
Token Inventory By Batch Report	
Token Batch Id	Specifies the token batch to be included. Multiple token batches can be included.
Token Model	Specifies the token model to be filtered.
Date Range of Token Batch Manufacture Time	Specifies the token batches to be filtered based on their manufacture time.
Token Batch To Be Expired Within (No. Of Months)	Specifies the token batches to be filtered based on how close they are to expiry.
Include Custom Attribute *	If specified as "Yes", the report will be filtered by the custom attributes defined within the token batch. To know more about where this can be configured, refer to Token Batch .
Token Status Display List *	If specified as "Filter by token status display list in global setting", the report will only include tokens with certain statuses as defined in the Token Status Display List attribute in the Token Management settings. To know more about how to configure this attribute, refer to Global Setting - Token Management .
Token Inventory By Group Report	
Token Group Id	Specifies the token group to be included in the report. Multiple token groups can be included.
Token Model	Specifies the token model to be filtered.
Token Status Display List *	If specified as "Filter by token status display list in global setting", the report will only include tokens with certain statuses as defined in the Token Status Display List attribute in the Token Management settings. To know more about how to configure this attribute, refer to Global Setting - Token Management .
Include Token Model Filtering	If specified as "No", token models will not be used as a filter in the report. If specified as "Yes", either of the following will apply depending on the value entered in the Token Model criteria. If a token model has been specified in the Token Model criteria, it will be used as a filter in the report.

Criteria	Description
	If the token model has not been specified, or if a wildcard was used in the Token Model criteria, the generated report will include all token models, along with their respective quantities. Default: Yes Note: When token models are not used as a filter, the report generation will be significantly faster.

Token Inventory Detail By Token Group Report

This report provides a detailed inventory of the tokens in the system. It will display the list of all token serial numbers, vendor, model, status, its assignee, assigned time and its assigner.

To locate the **Token Inventory Detail By Token Group** page, navigate to **Report Dashboard > Token & SMS > Real Time** and click **Token Inventory Detail By Token Group** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects the token serial number, token vendor, token model, token status, assignee Id, assignee name, assigned time and assigner Id. For an example, refer to [Token Inventory Detail By Token Group Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute	Description
Token Serial Number	The identification number assigned to the token.
Token Vendor	The vendor of the token.
Token Model	The token model of the token.
Token Status	Status of the token.
Assignee Id@User Store	The Id and user store of the assignee.
Assignee Name	The name of the assignee.
Assigned Time	The date and time the token is assigned.
Assigner Id@User Store	The Id and user store of the assigner.

Token Assignment Report

This report provides the list of tokens assigned to the assignee Ids/Token Sharing Group Ids.

To navigate to the **Token Assignment** page, select **Report Dashboard > Token & SMS > Real Time** and click **Token Assignment** on the left pane. Enter the relevant search criteria to generate the report. The generated report reflects the assignee Id or token sharing group Id, assignee name or assignee Id, token serial number, token vendor, token model, token status, assigned time, and assigner Id. For an example, refer to [Token Assignment Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute	Description
Assignee Id/Token Sharing Group ID@User Store	The Id of the assignee or the Id and user store of the token sharing group.
Assignee Name/(User Store)Assignee Id	The name or the Id of the assignee.
Token Serial Number	The identification number assigned to the token.
Token Vendor	The vendor of the token.
Token Model	The token model of the token.
Token Status	Status of the token.
Assigned Time	The date and time the token is assigned.
Assigner Id@User Store	The Id and user store of the assigner.

Token Expiry Report

This report provides the expiry time details of all tokens under the selected token group Id(s).

To locate the **Token Expiry** page, navigate to **Report Dashboard > Token & SMS > Real Time** and click **Token Expiry** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects the token expiry time and other token information. For an example, refer to [Token Expiry Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute	Description
Expiry Time	The expiry time of the token(s).
Manufacture Time	The manufacturing time of the token(s).
Token Batch Id	The Id of the token batch.
Token Vendor	The vendor of the token.
Token Batch Name	The name of the token batch.
Token Model	The token model of the token(s).
Token Policy	The token policy of the token(s).
Token Group Id	The Id of the token group.
Token Status	Status of the token(s).
Assignment State	Indicates whether the assignment state is assigned or unassigned.
Token Serial Number From	Indicates the first token serial number in the specified token group Id.
Token Serial Number To	Indicates the last token serial number in the specified token group Id.
Static Vector	The static vector of the token. Note: This is only used by VASCO Multi Device.
Token Serial Number	List of the imported token serial number(s).

Historical Reports

Historical reports allow the administrator to analyze the past token and SMS events. Below are the token and SMS related historical reports.

Token Activities Report

This report lists down all token events/activities based on the specified date/time.

To locate the **Token Activities Report** page, navigate to **Report Dashboard > Token & SMS > Historical** and click **Token Activities** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects the event timestamp, event type, token-related information and event result. For an example, refer to [Token Activities Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute	Description
Audit Id (Server Id/Actor User Id@User Store)	Indicates the audit Id or user Id and user store of the actor.
Event Timestamp	The date and time when the event occurred.
Event	Indicates the event type.
Batch/Group/Token/User	The name of the batch, group, token or user.
Token Serial Number(s)/Digital Signature Transaction Data	Indicates the token serial number(s) or digital signature transaction data.
Token Count/Token Status/Token Sharing Group ID	Indicates the token count, token status and token sharing group Id.
Event Result/Message/Result Code	Indicates the event result, message or result code.
Device Friendly Name/Device Instance Number/Device Type	Indicates the friendly name, instance number and device type of the device.

SMS OTP Usage Report

This report provides the number of SMS OTP usage per user Id.

To locate the **SMS OTP Usage Report** page, navigate to **Report Dashboard > Token & SMS > Historical** and click **SMS OTP Usage** on the left pane. Enter relevant search criteria to generate the report. The generated report reflects the user Id, user name, mobile number, email, and no. of SMS. For an example, refer to [SMS OTP Usage Report Sample](#).

You may refer to the following attribute descriptions in the generated report:

Attribute	Description
No.	Sequential number.
Actor User Id@User Store	The user Id and user store of the actor.
User Segment Id@Segment Store	The segment Id and user store of the user.
User name	The user name of the actor.
Mobile Number	Indicates the mobile number of the user.
Email	Indicates the email of the user.

Attribute	Description
No. Of SMS	Indicates the no. of SMS generated per user.

4. Administration for End-to-End Encryption Authentication

AccessMatrix End-to-End Encryption Authentication Solution (E2EEA) implements an end-to-end application layer encryption to protect PINs and other sensitive data in communications between terminals and hosts. E2EE usually refers to End-to-End Password Protection for security-sensitive applications such as internet banking, etc.

The E2EEA solution provides the following features to address the end-to-end protection requirements:

- **Integration with HSM**

E2EEA creates a secured channel between the client's access device and a Hardware Security Module (HSM). Within this channel, the Password is encrypted at the client's access device. It can only be decrypted for verification by the HSM located in a physically secure location within the organization. In doing so, the password and other sensitive data can never be exposed, not even to the organization's applications and servers.

The E2EEA solution supports most leading HSMs available in the market, and the following operations can be performed within the HSM:

- Encryption/Decryption
- Hash
- Protection of master encryption key
- Random number generation
- Initial pin generation
- PIN verification

- **Pseudo E2EE Authentication**

Although E2EE with HSM is recommended for deployment where HSM is unavailable, UAS offers a Pseudo E2EEA solution where a password is encrypted end-to-end from a client device to UAS, and verification is done in UAS.

- **PIN Mailer**

The E2EE PIN Mailer solution is a versatile authentication server that enables organizations to deliver sensitive documents to end-users, simplifying integration complexities. Refer to the *UAS PDF Pin Mailer Implementation Guide* for more information.

5. Administration for Contextual Authentication

Contextual Authentication is a server-side component of UAS that provides non-static authentication which is focused on providing dynamic security to enhance usability for users and strengthen security based on configured policies. It will take into account the profile of the user requesting access to the system to determine the risk profile associated with the login. The risk profile is then used to determine the complexity of the challenge. Higher risk profiles lead to stronger challenges, whereas one-factor authentication may suffice for lower-risk profiles. This risk-based implementation allows the application to challenge the user for additional credentials only when the risk level is appropriate. You need to configure the contextual policy when you are administering the authentication realm and increase the risk score to apply the contextual policy of the realm for remote IP access as required.

Features

In AccessMatrix UAS, Contextual Policy is used to compute the risk score of each login and enforce relevant action upon a user requesting access to the system based on the following elements:

Elements	Description
Location	Risk score can be configured and associated with certain countries that the organization may deem uncommon to be the originating countries of the user access request. Risk categories of the country are defined as High Risk, Mid Risk, and Low Risk. The user's country would be determined based on 3rd party Geolocation database, given the user's contextual data, or either IP Address* or Location Latitude/Longitude*. <i>* - Mobile Internet banking application scope to provide information.</i>
Time	This is based on a two-time range (i.e. Business and non-business hours). Non-business hours would be attributed to a certain amount of risk score.
Device	User's device with rooted or jail-broken status could be attributed to a certain risk score. A new device login can also be attributed to an increased risk score up to a certain amount. The new device can be a mobile device, token or web browser. For now, the risk score increases for a new browser\device. There is no limit to the number of devices a user can use.
Historical Geolocation	Increase user risk score if the user requesting access came from a new country, a new subdivision or a new city.
Historical Geolocation Velocity	The velocity and risk score values can be configured to be higher depending on the login at different locations.

Elements	Description
Remote IP	AccessMatrix UAS supports IP-based authentication to provide restriction and reject a request if the user's IP is outside a specific IP range. If a user's IP is within a specified IP range, UAS will enable 1FA or 2FA authentication. A user's IP can be in either IPv4 or IPv6 format. You can specify one or multiple IP\IP ranges in the contextual policy of the realm for remote address risk assessment.
User Agent	The User-Agent is a standard HTTP header attribute that identifies the characteristics of the client-side application, such as the user's web browser, operating system, and device. You can configure the contextual authentication policy to check if the User-Agent string of the user's client-side application contains a match with the regular expression value specified in the policy.
User Group	You can configure different risk score values for different user groups. Use this if you intend to organize user groups according to varying risk factors.



Note: The risk score is computed after the credential verification of each login.

Supported 3rd Party Geolocation DB for Country Resolution

Geolocation Database	Description
IP-based on MaxMind (GeoIP2) DBs	- File based in Binary MDDb format (https://www.maxmind.com/en/geoip2-city) - Additional 3rd License apply - Supported in GeoIP2 binary MMDB format
Latitude/Longitude-based (GeoNames) DB	- File-based Geographical Database downloadable from (http://download.geonames.org/export/dump/) - Need to buy commercial version for support and accuracy.



Note: If other 3rd Party DBs are required, it would not be supported out of the box as additional customization might be required.

Download and Setup

The geolocation database is not included with UAS product release. Obtain the database file and place it to *res/* folder of AccessMatrix directory.

For example:

- Sample IP-based database (MaxMind) - <http://dev.maxmind.com/geoip/geoip2/geolite2/>
E.g. C:\Program Files(x86)\AccesMatrix\tomcat\res\GeoLite2-city.mmdb

-
- Sample Geolocation DB (Geonames) - <http://download.geonames.org/export/dump/>
E.g. C:\Program Files(x86)\AccessMatrix\tomcat\res\geonames-custom.txt

These databases are for testing use. For production use, read [Supported 3rd Party Geolocation DB for Country Resolution](#) section.

**Notes:**

- Geolocation DBs are flat binary files that need to be updated by customer leveraging on their existing operational processes for file updates.
- Updates of DB requires AccessMatrix server restart.
- Accuracy is fully dependent on the elements provided and information available within the Geolocation DB and we are not liable for any inaccuracies.

Configure Contextual Policy



Note: Before doing any configuration in the Admin Console, secure the server license required for Contextual Authentication support. Ensure that **Contextual Authentication** license feature is set to "Yes" to enable Contextual Authentication support and display the **Contextual Policy** menu.

This policy is optional during authentication realm administration. To access this page, navigate to **Administration Dashboard > Security Policy > Authentication > Contextual Policy**.

Contextual Policy	Description
Score-Based Contextual Authentication Policy	This policy contains the risk score assessment depending on the country, geolocation, time-of-day and rooted status. This also defines the contextual actions to be done based on the calculated risk score. By default, a predefined contextual policy called "DefaultScoreBasedContextualAuthnPolicy" will be created upon successful installation of AccessMatrix.
Rule-Based Contextual Authentication Policy	This policy allows configuration of assessment criteria based on risk countries, geolocation historical risk, time of day risk, rooted status risk, malware detection, and bot or automated transaction as available rules.
Script-Based Contextual Authentication Policy	This policy allows for the configuration of assessment criteria using scripted expressions based on risk countries, geolocation historical risk, time of day risk, rooted status risk, malware detection, and bot or automated transaction as available rules. Multiple conditions can be configured for a single rule.

The **Contextual Policy** page displays the list of policies that can be managed. From this page, the following actions can be performed:

- **Create** a contextual policy - Click the **Create** button to choose the type of implementation. Click the **Next** button to display the **Create Contextual Policy** page of the selected policy. Save the changes.
- **Search** for a contextual policy - Enter a contextual policy Id and/or contextual policy name and click the **Search** button.
- **View** a contextual policy - Click on a contextual policy Id under **Search Contextual Policy Result** to display the **View Contextual Policy** page.
- **Edit** a contextual policy - Click the **Edit** button from the **View Contextual Policy** page and modify the attribute(s) to change. Save the changes.
- **Delete** a contextual policy - Click the **Delete** button from the **View Contextual Policy** page. Click **OK** on the confirmation pop-up to delete the policy.

Configure Score-Based Contextual Policy

Create Score-Based Contextual Policy Attribute Descriptions

i Note: '*' denotes a mandatory attribute.	
Attribute Name	Description
*Contextual Policy Id	Identification assigned to the contextual policy.
*Contextual Policy Name	The name of the contextual policy.
Description	Brief description for the contextual policy.
Risk Scoring Tab	
Initial Risk Score	Initial risk score value. Value should be from 0 to 100. Default: 0
If Geolocation cannot be determined	Specify the default risk country value if geolocation cannot be determined. Options: • "treat as High Risk Country" • "treat as Mid Risk Country" • "treat as Low Risk Country" Default: treat as High Risk Country

Attribute Name	Description
Low Risk Countries	
Countries	This defines the list of countries that have low risk.
Increase Risk Score by (0~100)	The risk score value to be added if the login user belongs to a low-risk country. Value should be from 0 to 100. Default: 0
Mid Risk Countries	
Countries	This defines the list of countries that have a medium risk.
Increase Risk Score by (0~100)	The risk score value to be added if the login user belongs to a mid risk country. Value should be from 0 to 100. Default: 30
High Risk Countries	
Countries	This defines the list of countries that have high risk. Default: All Others
Increase Risk Score by (0~100)	The risk score value to be added if the login user belongs to a high-risk country. Value should be from 0 to 100. Default: 100
Geolocation Historical Risk Setting	
Maximum number of previous successful login to monitor	Define the maximum number of previous successful login to monitor. Value should be from 0 to 100. Default: 10
If Country is "new", increase risk score by	Specify the increase in risk score if the country is new. Value should be from 0 to 100. Default: 40
If Subdivision is "new", increase risk score by	Specify the increase in risk score if the subdivision is new. Value should be from 0 to 100. Default: 30
If City is "new", increase risk score by	Specify the increase in risk score if the city is new. Value should be from 0 to 100. Default: 20
Historical Geolocation Velocity Risk Setting	

Attribute Name	Description
Based on latitude and longitude history, if velocity is faster than (km/h)	Specify the velocity value in km/h based on the latitude and longitude history of login at different locations. For example: "5" (walking) "15" (running) "40" (cycling) "150" (car) "400" (train) "1000" (plane) "2000" (concorde) Please ensure that the Maximum number of previous successful login to monitor field is properly set.
Increase Risk Score by (0~100)	Default: 20
Time-of-day Risk Assessment	
Enabled?	If set to "Yes", time risk assessment will be enabled. The risk scoring will be based on the user's local time, which is derived from the server time and the user's time zone. The user's time zone will be determined using their current geolocation data. Note: The geolocation database is required to implement this risk scoring. If the time zone cannot be determined, this risk scoring will be skipped. Default: Yes
If Time-of-day from	Start time. Default: 18:00
to	End time. Default: 09:00
Increase Risk Score by (0~100)	The risk score value to be added if the user logs in within the specified time. Value should be from 0 to 100. Default: 30
Server Time-of-day Risk Assessment	
Enabled?	If set to "Yes", server time risk assessment will be enabled. The risk scoring will be based on the user's login time, which is calculated using the server time. Default: Yes

Attribute Name	Description
If Time-of-day from	Start time. Default: 18:00
to	End time. Default: 09:00
Increase Risk Score by (0~100)	The risk score value to be added if the user logs in within the specified time. Value should be from 0 to 100. Default: 30
Rooted Status Risk Assessment	
Enabled?	Setting to "Yes" will enable rooted status risk assessment. Default: Yes
If device is rooted/jailbroken	This defines the risk score value when the device is rooted/jailbroken. Options: • "(default)" • "Increase risk score by the value in the next field" - Increase the existing risk score to the value provided. • "Set minimum risk score to the value in the next field" - Set the value as the minimum risk score if the existing score is lower than the defined score value. Default: Set minimum risk score to the value in the next field
	Text Field: Specify the score value corresponding to the If device is rooted/jailbroken field. Value should be from 0 to 100. Default score value: 30
New Device Risk Assessment	
Enabled?	Select "Yes" or remain as default will enable new browser\device risk assessment. Default: Yes
If device is "new", increase risk score by	The risk score value to be added if the browser\device is new. The value should be from 0 to 100. Default: 20
Remote Address Risk Assessment	
Enabled?	Select "Yes" or remain as default will enable remote IP address risk assessment for multiple specified IP ranges. Default: Yes

Attribute Name	Description
If IP is within this range	Specify one or multiple IP ranges to restrict users' IP-based accesses at remote. Each entry can either be the exact IP address or ranges in either IPv4 or IPv6 format. For example: • Exact IP address - "10.1.2.3" or "fe80::5dd4:37ab:c807:8af1" • Range - "10.1.2.100-10.1.2.200" or "fe80::5dd4:37ab:c807:8af1-fe80::5dd4:37ab:c807:8fff" Note: The dash "-" denotes "to" in an IP range.
Increase Risk Score by (0~100)	The risk score value to be added if a remote user's IP address is within the IP ranges. The value should be from 0 to 100. Default: 20
User Agent Risk Assessment	
Enabled?	Set to "Yes" to enable user agent matching. Default: Yes
User agent matches	Specify the value using a regular expression such as ".*Chrome.*". If the User-Agent string contains an instance of the specified value (e.g. "Chrome"), it is considered a match.
If user agent matches, increase risk score by	Specify how much the risk score should be increased when the User-Agent string contains a match with the specified value. Value should be from 0 to 100 Default: 20
User Group Risk Assessment	
Enabled?	Set to "Yes" to enable user group matching. Default: Yes
User is a member of	Select the user group id(s) of the user group(s) that will be matched for risk assessment.
If User is a member of group, increase risk score by	Specify how much the risk score should be increased when the user is a member of the user group(s) specified under the User is a member of attribute. Value should be from 0 to 100 Default: 20
Contextual Action Tab	

Attribute Name	Description
This tab contains the action to be performed based on the calculated risk score value. The administrator can define the action, e.g. "Deny Login", and assign the respective login modules to use based on the calculated risk score.	
If Risk Score more than	Specify the risk score value. The value should be from 0 to 100. Default: 80
	Drop-down list: Specify the number of login modules to use in relation to the If Risk Score more than attribute. Login modules must be configured in the Authentication Realm. Options: • "(default)" • "1FA only" • "1FA+2FA" • "1FA+2FA+3FA" • "1FA+2FA+3FA+4FA" • "All FA" • "Deny Login" Default: Deny Login
else if Risk Score more than	Specify the risk score value. The value should be from 0 to 100. Default: 20
	Drop-down list: Specify the number of login modules to use in relation to the else if Risk Score more than attribute. Login modules must be configured in Authentication Realm. Options: • "(default)" • "1FA only" • "1FA+2FA" • "1FA+2FA+3FA" • "1FA+2FA+3FA+4FA" • "All FA" • "Deny Login" Default: 1FA+2FA
else	If Risk Score is lower than the values specified for the previous two attributes.
	Drop-down list: Specify the number of login modules to use in relation to the else attribute. Login modules must be configured in Authentication Realm. Options:

Attribute Name	Description
	"(default)" "1FA only" "1FA+2FA" "1FA+2FA+3FA" "1FA+2FA+3FA+4FA" "All FA" "Deny Login" Default: 1FA only



Notes:

- The **Countries** field should use the ISO 3166-1 Alpha-2 Country Code values as shown in [B. ISO 3166-1 Alpha-2 Country Codes](#).
- Total Risk Score is calculated based on the defined settings in the **Risk Scoring** tab.

Configure Rule-Based Contextual Policy

Create Rule-Based Contextual Policy Attribute Descriptions



Note: '*' denotes a mandatory attribute.

Attribute Name	Description
*Contextual Policy Id	Identification assigned to the contextual policy.
*Contextual Policy Name	The name of the contextual policy.
Description	Brief description for the contextual policy.
Attributes Tab	
Geolocation Historical Risk Setting	
Maximum number of previous successful login to monitor	Defines the maximum number of previous successful login to monitor. Value should be from 0 to 100. Default: 10
Rule-Based Contextual Action Tab	
This tab contains the action to be performed based on certain rules. The following columns will be displayed: Rule, Rule Value, Rule-Based Action, Event Notification Policy (optional), and Action.	

Adding Rules for Rule-Based Contextual Policy



Note: The default rule will only be executed if none of the added rules is satisfied. Its rule-based action by default is "perform1FA". Select the **Edit** button to change its action as required.

1. Click the **Add** button. The **Add New Rule Based Action** dialog box will be displayed.
2. Select a **Rule**. Below is the list of options:

Rule	Rule Value
If Geolocation cannot be determined	N/A
If within this country list	Specifies the country's ISO code, e.g. "SG,CN,MY" for Singapore, China and Malaysia.
Based on Geolocation login history, if the country is new	N/A
Based on Geolocation login history, if the subdivision is new	N/A
Based on Geolocation login history, if the city is new	N/A
Based on latitude and longitude history, if velocity is faster than	Specifies the velocity value in km/h based on the latitude and longitude history of login at different locations. For example: ○ "5" (walking) ○ "15" (running) ○ "40" (cycling) ○ "150" (car) ○ "400" (train) ○ "1000" (plane) ○ "2000" (concorde)
If within these times of the day	Specifies the login time period allowed for users, e.g. "17:00-09:00". This rule will be based on the user's local time, which is derived from the server time and the user's time zone. The user's time zone will be determined using their current geolocation data.

Rule	Rule Value
	Note: The geolocation database is required to implement this rule. If the time zone cannot be determined, this rule will be skipped.
If within these server based times of the day	Specifies the login time period allowed for users, e.g. "17:00-09:00". The user's login time is calculated using the server time.
If user's device is rooted	N/A
If malware is detected	N/A
If bot or automated transaction is detected	N/A
If device is new	N/A
If within these IP ranges	Specifies the IP ranges of IPv4/IPv6 using a "-" (dash) separator such as "192.168.1.0-192.168.1.255", or just a single IP such as "192.168.1.45". For multiple values or ranges, use comma delimiter.
If user agent matches	Specifies the value using a regular expression such as ".*Chrome.*". If the User-Agent string contains an instance of the specified value (e.g. "Chrome"), it is considered a match.
If user is a member of group(s)	User can click Select Group(s) and select the desired user group id(s) to be matched for risk assessment. The appropriate user store containing the user group(s) should first be selected for the user group(s) to be displayed.

3. Select an **Action**. Below is the list of options:

Action	Event Notification Policy
1FA only	N/A
1FA + 2FA	N/A
1FA + 2FA + 3FA	N/A
1FA + 2FA + 3FA + 4FA	N/A
All FA	N/A
Deny Login	N/A
Send Notification	User can specify the notification policy to use.

4. Click the **Apply** button to add the new rule-based action. Otherwise, click the **Cancel** button not to add a new rule-based action.

-
- Repeat steps 1-4 to add a new rule again.

Deleting Rules for Rule-Based Contextual Policy

From the list of rule-based contextual actions, click the **Delete** button of the respective rule to delete.

Configure Script-Based Contextual Policy

Create Script-Based Contextual Policy Attribute Descriptions

i Note: '*' denotes a mandatory attribute.	
Attribute Name	Description
*Contextual Policy Id	Identification assigned to the contextual policy.
*Contextual Policy Name	The name of the contextual policy.
Description	Brief description for the contextual policy.
Attributes Tab	
Maximum number of previous successful login to monitor	Defines the maximum number of previous successful logins to monitor. Value should be from 0 to 100. Default: 10
Script-Based Contextual Action Tab	
This tab contains the actions to be performed based on certain rules. The following columns will be displayed: Rule , Condition(s) , Rule-Based Action , and Event Notification Policy (optional) .	

Adding Rules for Script-Based Contextual Policy

i Note: The default rule will only be executed if none of the added rules is satisfied. Select the Edit button to change its action as required. This rule cannot be deleted.
--

- Under the **Script-Based Contextual Action** tab, click **Add** button. The **Add Rule** window will be displayed.
 - Enter a **New Rule Name**.
 - Enter the condition(s) for this new rule by entering a scripted expression into the text box using the supported expressions and logical operators.
The following is an overview of the supported expressions:
-

Expression	Corresponding Rule	Rule Value
context.isGeoLocationNotDetermined()	If Geolocation cannot be determined	N/A
context.isWithinCountriesList(String countries)	If within this country list	User can specify the country's ISO code, e.g. "context.isWithinCountriesList("SG")", "context.isWithinCountriesList("MY,CN")"
context.isCountryNew()	Based on Geolocation login history, if the country is new	N/A
context.isSubdivisionNew()	Based on Geolocation login history, if the subdivision is new	N/A
context.isCityNew()	Based on Geolocation login history, if the city is new	N/A
context.isVelocityFasterThan(double velocityLimitInKmh)	Based on latitude and longitude history, if velocity is faster than	User can specify the velocity value in km/h, e.g. "context.isVelocityFasterThan(100)"
context.isWithinTimeOfDay(String timeOfDayFrom, String timeOfDayTo)	If within these times of the day	Specifies the login time period allowed for users, e.g. "context.isWithinTimeOfDay("18:01", "07:59")". This rule will be based on the user's local time, which is derived from the server time and the user's time zone. The user's time zone will be determined using their current geolocation data.

Expression	Corresponding Rule	Rule Value
		Note: The geolocation database is required to implement this rule. If the time zone cannot be determined, this rule will be skipped.
context.isWithinServerTimeOfDay	If within these server based times of the day	Specifies the login time period allowed for users, e.g. "context.isWithinServerTimeOfDay("18:01","07:59")". The user's login time is calculated using the server time.
context.isRootedStatus()	If user's device is rooted	N/A
context.isMalwareDetected()	If malware is detected	N/A
context.isBotOrAutoTrx()	If bot or automated transaction is detected	N/A
context.isNewDevice()	If device is new	N/A
context.isIPAddressWithin(String strRanges)	If within these IP ranges	User can set the IP ranges of IPv4/IPv6 as such: - No separator for single IP address, e.g. "context.isIPAddressWithin("192.168.50.1")" - Dash separator for range of IP addresses, e.g. "context.isIPAddressWithin("192.168.50.1-192.168.50.255")" - Comma delimiter for multiple ranges of IP addresses, e.g. "context.isIPAddressWithin("192.168.50.1-192.168.50.255, 192.168.1.1-192.168.1.255")"
context.isUserAgentMatch(String regExp)	If user agent matches	User can specify a value using a regular expression such as ".*Chrome.*", e.g. "context.isUserAgentMatch(".*Chrome.*")" If the User-Agent string contains an instance of the specified value (e.g. "Chrome"), it is considered a match.

Expression	Corresponding Rule	Rule Value
context.isUserGroupMatch(String storeGroupId)	If user group matches	User can specify the desired user group id(s) to be matched for risk assessment as such: - Specify the user store of the desired user group before the user group, e.g. "context.isUserGroupMatch("DefaultStore:InitialAdminGroup")" - Comma delimiter for multiple user groups, e.g. "context.isUserGroupMatch("DefaultStore:InitialAdminGroup, ActiveDirectory:i-Sprint")"

Additionally, the following are the supported logical operators:

Operator	Description
!	NOT
&&	AND
	OR
()	Groups expressions together

For example, to create a rule where access is denied if the user's device is rooted AND the user's login time is NOT within office hours, enter "context.isRootedStatus() && !

"context.isWithinTimeOfDay("08:00", "19:00")" into the text box and select "Deny Login" for the action.

4. Select an action. Below is the list of action options:

Action
1FA only
1FA + 2FA
1FA + 2FA + 3FA
1FA + 2FA + 3FA + 4FA
All FA
Deny Login

5. Select the event notification policy to use, if applicable.
6. Click the **Apply** button to add the new rule-based action. Otherwise, click the **Cancel** button to not add a new rule-based action.
7. Repeat steps 1-6 to add a new rule.
8. Click **Save** on the **Create Contextual Policy** page.

Adding Custom Rules for Script-Based Contextual Policy

In addition to the supported expressions described above, it is possible to create custom contextual authentication rules using user-defined expressions. To do so, the following requirements must be met:

1. When the application calls the AccessMatrix Login API, the information required for the custom rules must be contained as variables of a `context.vars` object in the request parameters.

The example below shows a sample request where transaction data (namely, `transactionType` and `transactionAmount`) is provided in the API call:

Sample Request
JSON
<pre>{ "userId": "demouser", "password": "password1", "params": { "realmId" : "40151", "context.vars": { "transactionType": "MoneyTransfer", "transactionAmount": 10000 } } }</pre>

2. When creating the custom rule in the Admin Console, the user-defined expression should adhere to the following formats:

Expression Format	Sample Expression	Corresponding Rule
<code>(<value>.equals(context.vars().get(<contextual variable>))</code> Note: Avoid using the format <code>"context.vars().get(<contextual</code>	<code>("MoneyTransfer").equals(context.vars().get("transactionType"))</code>	If the specified contextual variable is equal to the specified

Expression Format	Sample Expression	Corresponding Rule
variable>).equals(<value>)", as an exception will be encountered if the contextual variable is not found within the API call.		value E.g. if transactionType equals "MoneyTransfer"
context.vars().getOrDefault(<contextual variable>, <default value>)<operator><value> Note: Avoid using the format "context.vars().get(<contextual variable>)<operator><value>", as an exception will be encountered if the contextual variable is not found within the API call.	context.vars().getOrDefault("transactionAmount",10001)<=10000	If the value corresponding to the specified contextual variable meets the condition stipulated by the specified operator and value Or, if no value is specified for that variable in the API call, use the specified default value to evaluate E.g. if transactionAmount is less than or equal to 10,000 Or, if no value for transactionAmount is specified in the API call, assume amount is 10,001 (i.e. greater than 10,000;

Expression Format	Sample Expression	Corresponding Rule
		condition is not met)

Replace the content enclosed within "<>" as appropriate:

- a. <contextual variable> represents a variable of the context.vars object in the API call (e.g. transactionType or transactionAmount). The specified variable name must match the name of the corresponding variable in the API call.

i **Note:** Custom rules also support arrays and nested JSON objects. In such cases, <contextual variable> should be replaced with a valid expression that contains the index/object name of the desired variable within the array/nested object (e.g. "context.vars().get("transactionList").get(0)" or "context.vars().get("transactionList").get("transfers)" respectively).

- b. <operator> represents a valid operator (e.g. "<", ">=", etc.).
- c. <value> represents a valid specified value for a contextual variable.
- d. <default value> represents the default value to use when evaluating the condition if no value is specified for the corresponding contextual variable in the API call. This value must be specified to account for such scenarios.

For example, with reference to the previous samples, to create a rule where 1FA will be performed if the transaction type is "MoneyTransfer" and the transaction amount is less than or equal to 10,000, enter

("MoneyTransfer").equals(context.vars().get("transactionType")) &&

context.vars().getOrDefault("transactionAmount",10001)<=10000" into the text box and select "1FA only" for the action.

Script behavior when variables are missing:

If any of the contextual variables or values specified within the custom rule are not found in the API call, the following behavior applies:

- For conditions using the format "(<value>).equals(context.vars().get(<contextual variable>))", the condition will be considered not satisfied.
- For conditions using the format "context.vars().getOrDefault(<contextual variable>, <default value>)<operator><value>", the condition will be evaluated using the <default value> provided.

The following example illustrates this behavior:

1. An API call is made with no variables under context.vars:

Sample Request (Missing variables)	
JSON	<pre>{ "userId": "demouser", "password": "password1", "params": { "realmId" : "40151", "context.vars": { } } }</pre>

2. A script-based contextual authentication policy is created with a rule, "Rule1":
 - a. **Script:** `("MoneyTransfer").equals(context.vars().get("transactionType")) || context.vars().getOrDefault("transactionAmount",10001)<=10000`
 - b. **Action:** "1FA only"
3. "Rule1" is evaluated with the following result:
 - a. **Condition 1:** `("MoneyTransfer").equals(context.vars().get("transactionType"))`
The variable `transactionType` is not found in the API call.
As there is no `transactionType` variable with the value "MoneyTransfer", the condition is not satisfied.
 - b. **Condition 2:** `context.vars().getOrDefault("transactionAmount",10001)<=10000`
The variable `transactionAmount` is not found in the API call. Thus, this condition is evaluated as if the parameter is provided with the default value specified, i.e. "10001".
As `transactionAmount` is *not* `<= 10000`, the condition is not satisfied.
 - c. **Final outcome:**
As the script uses the `"||"` logical operator, only one of the conditions must be met for the rule to be satisfied. As both conditions are not met, the rule is not satisfied.
The action "1FA only" is not performed.

i	Note: Refer to the AccessMatrix REST API Specification and UAS Contextual Authentication Developer Guide - UAS Contextual Authentication Implementation Using API for more information on implementing contextual authentication via REST API.
----------	---

Configuring Rule Priority for Script-Based Contextual Policy

If multiple rules are configured, the rules are assessed from top to bottom. If a rule is satisfied, the corresponding action will be applied and no more rules will be assessed. If none of the configured rules are satisfied, the action assigned to the default rule is applied.

The rule priority can be configured using the **Up** and **Down** buttons on the respective rules. Clicking **Up** increases the priority of the corresponding rule; clicking **Down** reduces the priority of the rule. The default rule cannot be moved.

Editing Rules for Script-Based Contextual Policy

Existing rules can be modified in the following ways:

- To edit an existing rule, click on the **Edit** button of the respective rule.
- To delete an existing rule, click on the **Delete** button of the respective rule.

6. Biometrics

Biometrics authentication solution is an all-in-one authentication system which can be integrated with the biometrics technologies such as the face, fingerprints, iris, voice, palm vein recognition, and so on. It supports a mobile app to fulfill face recognition and authentication for desired functions such as login and transaction authorization. It also provides a flexible platform for all biometric authentication providers and it can easily support new technology using PAM technology. It also supports remote account opening by adopting a smooth and high-quality video, stable and secured communication real-time face capture and a leading face recognition technology with full lifecycle monitoring.

The face recognition process is divided into 5 parts: face detection, aliveness detection, face image capture, preprocessing and characteristics extraction.

- **Face Detection**

During the initial face recognition process, an appropriate human face has to be detected by detecting from different angles and a user's natural facial movements.

- **Aliveness Detection**

It can prevent identity fraudulence by effectively distinguishing the non-aliveness images such as the printed photos, the photos shown on the mobile phone or pad, and video from the real human images.

Biometrics solution provides Authentication Logs management. It records the information of each transaction that is authenticated by face recognition. Every transaction that is authenticated by face recognition can be traced back in the system logs. The system supports export logs partly based on filter requirements or export all logs for future audit.

Biometric Policy

A biometric policy is where a verification threshold for score matching evaluation is being set. Navigate to **Administration Dashboard > Security Policy > Biometric Policy**. From this page, the following actions can be performed:

- **Create** a biometric policy - Click the **Create** button and choose the "Basic Biometric Policy Handler" implementation. Click the **Next** button to display the **Create Biometric Policy** page. Provide the necessary information and save the changes.

- **Search** for a biometric policy - Enter a biometric policy Id and/or biometric policy name as search criteria and click the **Search** button.
- **View** a biometric policy - Click on a biometric policy Id under **Search Biometric Policy Result** to display the **View Biometric Policy** page.
- **Edit** a biometric policy - Click the **Edit** button from the **View Biometric Policy** page and modify the attribute(s) to change. Save the changes.
- **Delete** a biometric policy - Click the **Delete** button from the **View Biometric Policy** page. Click **OK** on the confirmation pop-up to delete the biometric policy.

Attribute Name	Description
*Biometric Policy Id	Identification assigned to the biometric policy.
*Biometric Policy Name	The name of the biometric policy.
Description	Brief description for the biometric policy.
Implementation	Type of biometric policy implementation, i.e. "Basic Biometric Policy Handler".
Attributes Tab	
*Threshold	Biometric verification threshold score. Any score above this threshold represent positive biometric match. Value should be from 0 to 100. Default: 80

The biometric policy is used in Biometric Vendor configuration (navigate to **Administration Dashboard > Configuration > Biometrics > Biometric Vendor**). The configuration attributes will depend on the biometric vendor plugin loaded in AccessMatrix server.

Example:

You are here: Configuration > Biometrics > Biometric Vendor

View Biometric Vendor

*Biometric Vendor Id:	FacePlusplus	*Biometric Vendor Name:	FacePlusplus
Description:		Version :	1
Implementation:	Face++ Face Biometric Handler	Biometric Vendor Id Integer:	31005
Configuration			
Attribute Name	Attribute Value	Remark	
Biometric Policy	DefaultBasicBiometricPolicy		
Face++ Server Address	http://192.168.94.129:9000/	Default: http://localhost:9000/	
Face++ 1:N Server Address	http://192.168.94.129:9090/	Default: http://localhost:9090/	



Note: The **Create** button in the Biometric Vendor configuration page will not work. A biometric vendor is automatically created once the biometric plugin is loaded.

7. Administration for RADIUS

AccessMatrix UAS has a built-in RADIUS Server to provide RADIUS support. Remote Authentication Dial-In User Service (RADIUS) is a protocol for carrying information related to authentication, authorization, and configuration between a Network Access Server that desires to authenticate its links and a shared authentication server. It is a simple client-server protocol that uses UDP (User Datagram Protocol) and shared secret.

Firewall Port Opening

Ensure the following network firewall ports are open for communication from one component to another before deployment and configuration.

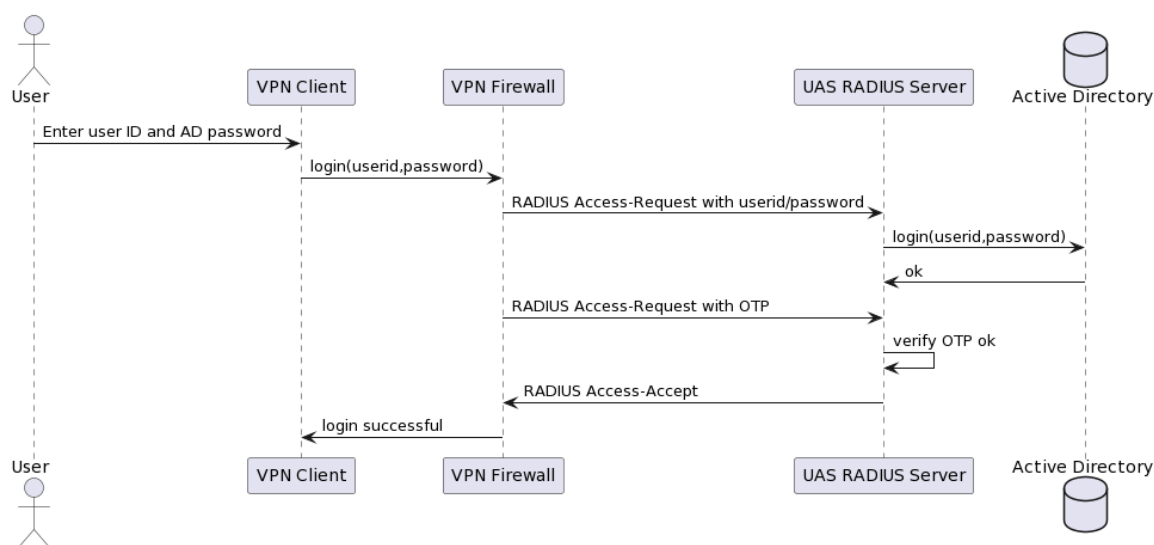
1. VPN Firewall to UAS RADIUS: UDP/1812 (RADIUS)
2. UAS RADIUS to RDBMS (Check **am5.xml** at `<ACMATRIX_ROOT>\tomcat\conf\Catalina\localhost\` for the port number of the database)
3. UAS RADIUS to SMS & Email Gateway (depends on the SMS gateway integration protocol)
4. Application / Admin Console to UAS server, for example: TCP/8080 (http), TCP/8443 (https)

AccessMatrix UAS Integration Use Cases with RADIUS

- **Authentication**

RADIUS is an authentication protocol for networking equipment, e.g. routers and firewalls to talk to an external authentication server.

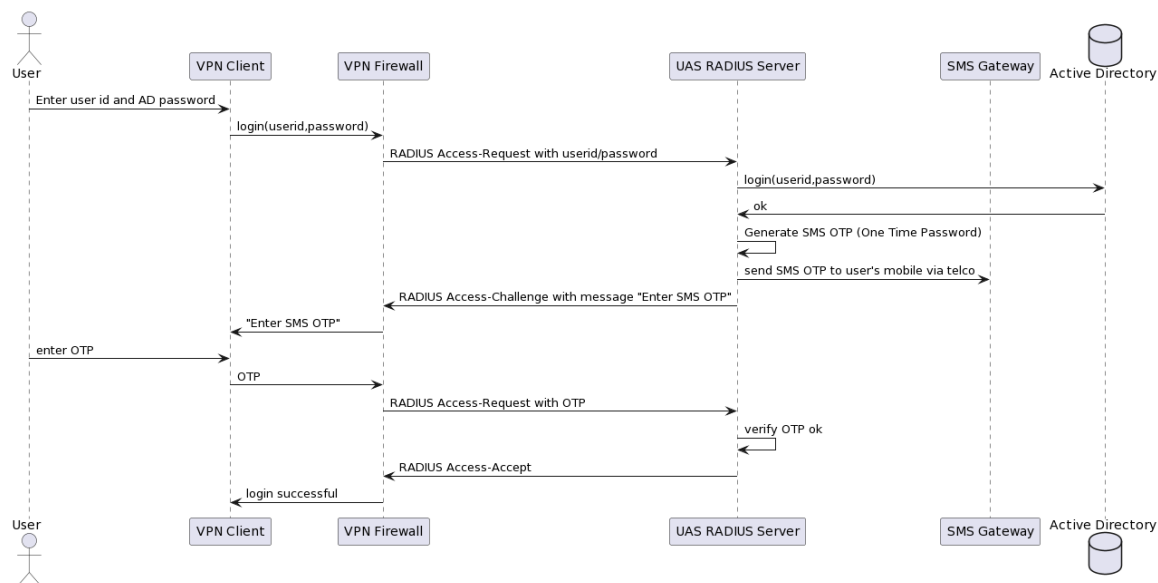
Below is a sample sequence diagram for AccessMatrix UAS 1FA with RADIUS-Compliant VPN:



- **Reauthentication**

In certain projects, an optional 2nd authentication based on SMS OTP or Token OTP is available. For example, all VPN users will authenticate using Microsoft AD password. However, to determine whether SMS OTP is required, checking must be performed against an LDAP group membership. If users belong to a group, 2nd authentication using SMS OTP is required.

Below is a sample sequence diagram for AccessMatrix UAS 2FA with RADIUS-Compliant VPN:



- **Act as Policy Information Point for Access Control**

Upon successful authentication, it is possible to return access control (policy) information (e.g. user's roles or groups), certain firewalls/VPN also use this to implement access control within the firewall/VPN devices. In this case, UAS server acts as PIP (Policy Information Point) and the NAS Client acts as PDP/PEP (Policy Decision & Enforcement Point).

- **Secure Temporary Guest Access to Enterprise Wireless LAN**

Guest access to wireless LAN using an OTP issued at the web portal. The actor will register his handphone number at a web application which calls UAS web service to generate OTP. A subsequent one-time password will be sent to the registered handphone and used by the guest to authenticate to wireless LAN.

Configure NAS Client

In the Admin Console, the NAS client specifies how the target devices or applications communicate with the UAS server for strong authentication using the RADIUS protocol.

To configure the NAS client, navigate to **Administration Dashboard > Configuration > Authentication > Enterprise 2FA > NAS Client** and click on the NAS client to be configured (e.g. "SampleNASClient"). On the **View NAS Client** page, click the **Edit** button.

The following table describes each of the configurable attributes:

Attribute Name	Description
*NAS Client Id	Identification assigned to the NAS client.
*NAS Client Name	The name of the NAS client.
Description	Brief description for the NAS client.
Version	The NAS Client revision. (uneditable)
Search filter for user (Segment)	Specifies the segment to which the user should be searched.
Configuration	
Connection	
The IP Address and Additional IP Address Ranges attribute values are not the end client's IP address but the IP address of the node that directly connects to the RADIUS server. For example, if there is a Reverse Proxy or Load Balancer in front of the AccessMatrix RADIUS server, then the IP address of the reverse proxy or load balancer must be specified.	
*IP Address	IP Address of the RADIUS client that initiates the RADIUS-based authentication request.
Additional IP Address Ranges	The IP Address range of the RADIUS clients. Each entry is either an exact address or a range. For example: <u>Exact address</u> • "10.1.2.3" <u>Range</u> • "10.1." means 10.1.0.0 to 10.1.255.255 • "10.1.2." means 10.1.2.0 to 10.1.2.255 • "10.1.2.100:10.1.2.200" means 10.1.2.100 to 10.1.2.200
*Secret	The secret passphrase shared between the RADIUS server and client.
Proxy RADIUS Realm Mapping	

Attribute Name	Description
This is used when performing a transparent pass through (i.e. radius proxy) to a third-party RADIUS server. Refer to Configure Transparent RADIUS Proxy for additional details.	
Proxy RADIUS Realm 1	The RADIUS realm of the third-party RADIUS authentication server. Append the "Proxy" suffix to the realm. For example, if the realm is "rsa", then enter "rsaProxy" as value.
Proxy Server IP for Proxy RADIUS Realm 1	IP Address of the third-party RADIUS authentication server.
Proxy Server Port for Proxy RADIUS Realm 1	Port number of the third-party RADIUS authentication server. Value is an integer with a value between 0-65535. Default: 1812
Proxy Server Secret for Proxy RADIUS Realm 1	Shared secret between the AccessMatrix server and the third-party RADIUS authentication server.
RADIUS Realm Mapping to AccessMatrix Authentication Realm	
Ignore Realm Separator (@)	Setting this attribute to "true" will ignore the "@" separator for realm selection, and the attribute value of Authentication Realm for User Id Without RADIUS Realm will always be used for authentication. It can be used when the User Id itself is an email address and "@" is not used for specifying realm. Default: false
*Authentication realm for User Id Without RADIUS Realm	If the RADIUS client sends a request with a User Id only, or the Ignore Realm Separator (@) attribute is set to "true", then this AccessMatrix authentication realm will be used.
Authentication Realm For User Id With RADIUS Realm (optional)	If the RADIUS client sends a request with a User Id and RADIUS realm, and the RADIUS realm matches a value specified in this attribute, then the AccessMatrix authentication realm corresponding to that value will be used for authentication. For example, if "rsa -> Active Directory Password (40172)" is added under this attribute, then a request with demouser@rsa will use the Active Directory Password (40172) authentication realm for login. Multiple authentication realms can be added under this attribute. Note: This attribute can only be used if Ignore Realm Separator (@) attribute is set to "false".

Attribute Name	Description
Group Bypass Authentication	
Enable Group Bypass Authentication	Set to "true" to skip user authentication if the user belongs to the specified group in User Group For Bypass Authentication attribute. Default: false
User Group For Bypass Authentication	Specifies the user group(s) to bypass and entries are separated using a semi-colon (";"). The user that belongs to any of these groups can submit an authentication request without password (i.e. no password verification).
2FA Reauthentication Based on Group Membership	
Enable 2FA Reauthentication Based on Group Membership	Set to "true" to enable 2FA reauthentication for the specified group in the User Group Id attribute. Refer to Configure 2FA Reauthentication section for a sample configuration. Default: false
User Group Id	The user group that requires 2FA reauthentication.
Login Module For 2FA Reauthentication	The login module to be used for 2FA reauthentication. Note: The user must be assigned with a realm containing this login module.
RADIUS Reply-Messages	
This is used to return a friendly message to the RADIUS client when authentication failed or challenged is required.	
Reply-Message For Access-Challenge	The message to be added in the RADIUS Access-Challenge Response. The user will receive a message with this format: <Reply-Message for Access-Challenge><space><Additional Challenge Message from AccessMatrix>. Example: If Reply-Message for Access-Challenge is set to "Authentication required.", then the user will receive "Authentication required. Second Authentication Factor is required." This message will be returned for these use cases: 1. 2FA Reauthentication 2. Wrong Password Default: Authentication required.
Reply-Message For Access-Reject	The message to be added in the RADIUS Access-Reject Response. The user will receive a message with this format: <Reply-Message for Access-Reject><space><Additional Challenge Message from AccessMatrix>.

Attribute Name	Description
	Example: If Reply-Message for Access-Reject is set to "Login failed.", then the user will receive "Login failed. (Object Not Found,0)." This message will be returned for these use cases: 1. If there is any exception thrown (i.e. Object not found or realm is not assigned) 2. Wrong Password Default: Login failed.
Send Access-Reject if Password is Wrong	Setting to "true" will send an Access-Reject response when password verification failed. Otherwise, Access-Challenge will be sent as RADIUS response. Default: false
Add RADIUS Response	
These attributes are used to return a RADIUS attribute to RADIUS client upon successful authentication. Refer to UAS for Access Control for additional details.	
RADIUS Response Class	Specifies the user response collector. The information collected will be added to the RADIUS response upon successful authentication.
RADIUS Response Attribute	Indicates the RADIUS attribute tag to be returned together with the user response. Options: • "Filter-Id (11)" - Indicates the name of the filter list for the user. • "Reply-Message (18)" - Indicates text that might be displayed to the user using the RADIUS server. • "Class (25)" - An arbitrary value that is available to be sent by the server to the client in an Access-Accept. Default: Filter Id (11)
RADIUS Response Format	Controls how the user response will be formatted as RADIUS attribute value. RADIUS response can be set based on the selected format. Example scenario: • "memberOf" and "userId" are configured in User Response Collector

Attribute Name	Description
	- user belongs to "Engineering" and "Employee" - userId = "user1" RADIUS Response Key Prefix attribute value is "UserAttr" - RADIUS response delimiter is semi-colon (;) In this scenario, the response will be displayed as follows: "Include both key and value (1)" Example response: memberOf=Engineering,Employee;userId=user1 "Include value only (2)" Example response: Engineering,Employee;user1 "Include value with single key prefix (3)" Example response: UserAttr=Engineering,Employee;user1 Default: Include value only (2)
RADIUS Response Key Prefix	Specifies the prefix value to be used for the "Include value with single key prefix (3)" option in RADIUS Response Format. Format: <RADIUS Response Key Prefix>[=]<User Response Value> Default: Group
RADIUS Response Delimiter	The delimiter to separate the user response attributes collected from the RADIUS Response Class. Default: ;
Vendor Specific Attributes(VSA)	
The method for communicating vendor-specific information between RADIUS client and RADIUS servers, allowing vendors to support their own extended attributes. Refer to Configure Vendor Specific Attributes (VSA) for sample configuration.	
Enable Vendor Specific Attributes	Set to "true" to include VSA entry in the RADIUS response. Default: false
VSA Response Class	Specifies the user response collector. Ensure that the user response must only have one attribute mapping as only the first attribute configured in the collector will be used as a mapping condition for VSA values.
Vendor Id	Indicates the Network Management Private Enterprise Number (PEN) assigned to the vendor. Refer to PEN List for the enterprise numbers assigned to vendors. Example: Fortinet Vendor ID = 12356

Attribute Name	Description
Vendor Specific Attributes Mapping	Mapping from user response value to VSA attribute value. Example: If the attribute "memberOf" has been set under VSA Response Class , and the AccessMatrix group (e.g. AMGroup1) has been mapped to a vendor group (i.e. AMGroup1=Group1), then "Group1" will be used as the VSA attribute value when an authenticated user belongs to the specified AccessMatrix group.
VSA Attribute Tag	Specifies the attribute tag as listed from the vendor's RADIUS VSA dictionary. The tag is an integer value. Example: 1 (attribute tag for Fortinet-Group-Name).
VSA Multiple Value Support	If this attribute is set to "true": - VSA attribute value will be considered as multivalued. - Each user response value will be searched in VSA Mapping. - The mapping result will be concatenated using VSA Multiple Value Delimiter attribute value. If this attribute is set to "false": - Only the first attribute value collected from VSA Response Class will be used to search in VSA mapping. Default: false
VSA Multiple Value Delimiter	The delimiter to separate multiple VSA attribute values. Default: ;
Others	
RADIUS Client Encoding	Specifies the character encoding of the RADIUS attributes value returned to the RADIUS client. Options: - BIG5 - CP2312 - GBK - GB2312 - ISO-8859-1 - UTF8 Default: UTF8
RADIUS Attribute Tag For Mapping Realm Id	Supports Juniper SSL-VPN to enable mapping from RADIUS attribute 44/32 to AccessMatrix authentication realm ID.
Force Change Password Support	If enabled, force change password will be handled with a proper message when the user's password is expired or force change password is required. Default: false

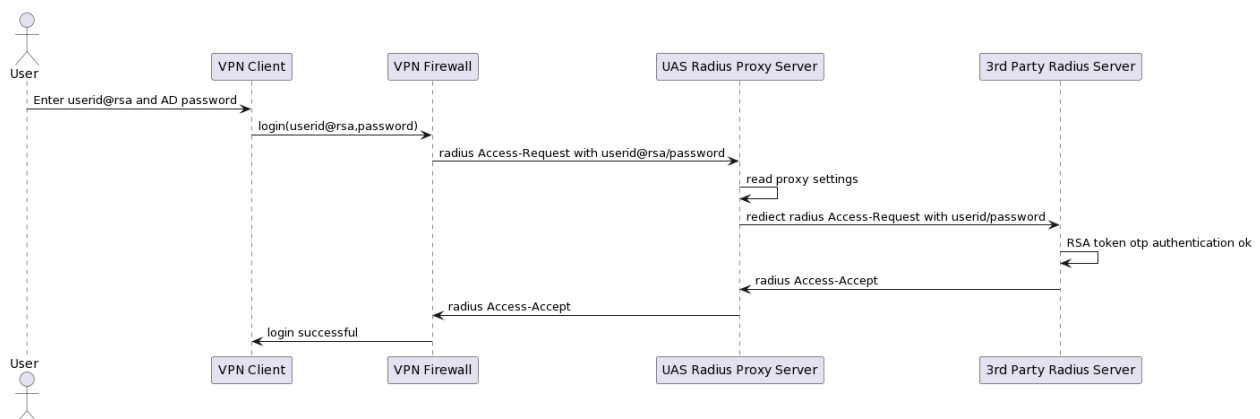
Attribute Name	Description
Deny Packets without Message-Authenticator	If set to "true", AccessMatrix will reject all RADIUS packets sent without the Message-Authenticator attribute. The Message-Authenticator is a checksum that is used to ensure the integrity of the RADIUS packet. This ensures that only requests sent with the Message-Authenticator attribute are processed, which mitigates the Blast-RADIUS vulnerability. Default: false Note: This attribute is set to "false" by default for backwards compatibility.

Configure Transparent RADIUS Proxy

In a certain project, it may be necessary to perform a transparent pass through (i.e. RADIUS proxy) to a 3rd party server such as RSA Authentication Server. UAS only supports one proxy per NAS client.

To use RADIUS proxy, configure the **Proxy RADIUS Realm Mapping** section under NAS client that requires the 3rd party RADIUS authentication server IP, port, RADIUS shared secret and the RADIUS realm. The RADIUS realm value is commonly appended to a user's user name and delimited with an "@" sign when a RADIUS client sends a request, e.g. "demouser@rsa" where "rsa" is the RADIUS realm. Append the "Proxy" suffix to the RADIUS realm (e.g. "rsaProxy") before assigning to the **Proxy RADIUS Realm 1** attribute.

Example User Id	Action
userId@rsaProxy	Strip off rsaProxy and forward to RSA authentication server.
userId(no realm)	Handled by authentication realm AD
userId@sms	Handled by authentication realm SMS



Configure 2FA Reauthentication

This feature enables rule-based 2nd authentication based on the group. An example of this is when a customer uses the Active Directory (AD) user store and require its employees to have 2FA (SMS OTP) after 1FA but will be excluding the managers for 2FA requirement.

To configure 2nd-factor Reauthentication:

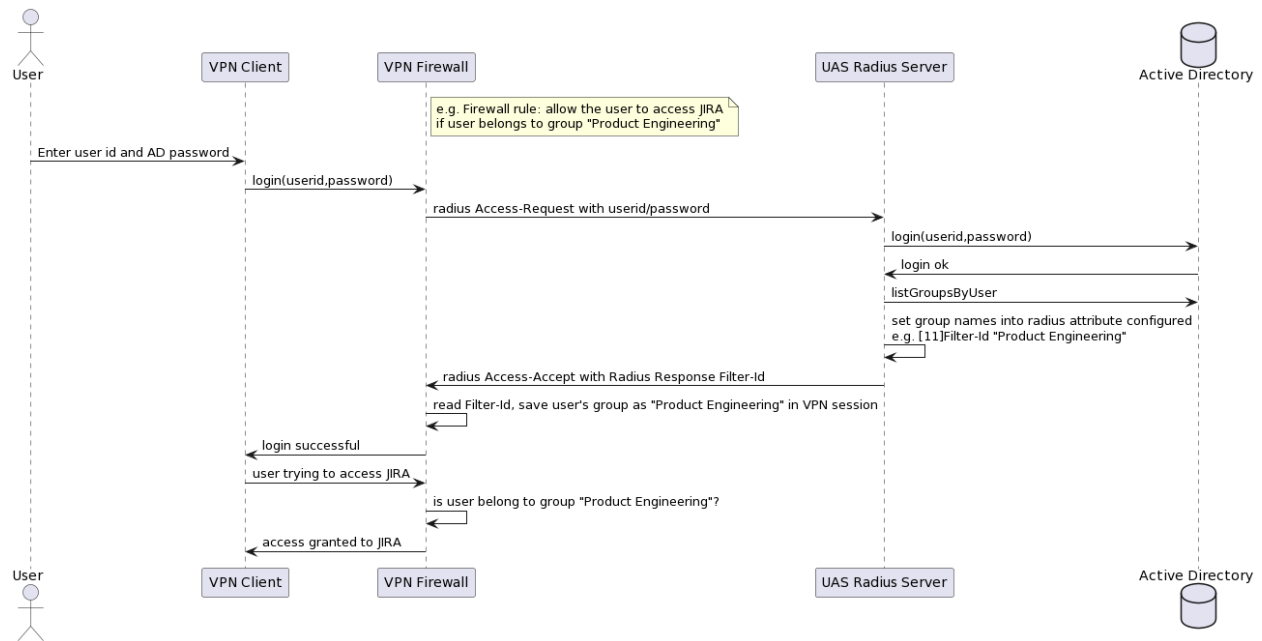
1. Configure in AD such that employees belong to a user group: "employees".
2. Configure the managers to join another group in AD with group Id: "managers".
3. Configure the rule-based 2nd-factor authentication in NAS Client:
 - Set the **Authentication Realm for User Id Without RADIUS Realm** to "40172".
 - Set **Enable 2FA Reauthentication Based on Group Membership** to "true".
 - Set the **User Group Id** to "employees".
 - Set the **Login Module for 2FA reauthentication** to "SMSOTPLLogin".
4. Save the changes.

UAS for Access Control

For some VPN firewalls, e.g. Juniper, it requires AD group members to be set in a specific RADIUS attribute (e.g. Filter Id) upon successful authentication. The group members will be used within its firewall rules for access control purpose. The following is a sample configuration:

1. Find the "**Add RADIUS Response**" section and make the following changes:
 - Set the **RADIUS Response Class** to "GroupMembership".
 - Set the **RADIUS Response Attribute** to "Filter Id(11)".
 - Set the **RADIUS Response Format** to "Include value only (2)".
 - Set **RADIUS Response Delimiter** to ";".
2. Save the changes.

After the configuration, the actual sequence would be like this:



Configure Vendor Specific Attributes (VSA)

Majority of VPN vendors support accepting user group membership for use in VPN access control. However, certain Network Access Server (NAS) like Huawei network devices may require certain vendor-specific attribute(VSA) to be set upon successful RADIUS authentication by the UAS for access control to take place within the switch/router.

Example 1:

To allow NAS Client to support VSA, e.g. Huawei-Exec-Privilege, enable the VSA in NAS Client configuration page by performing the following steps:

1. Set **Enable Vendor-Specific Attributes** to "true" to enable the feature.
2. Choose the appropriate **VSA Response Class**.
3. Set the **Vendor Id**, e.g. "2011" for Huawei.
4. Set the key-value pair to **Vendor-Specific Attributes Mapping** attribute.
5. Set the **VSA Attribute Tag**, e.g. "29" (Huawei-Exec-Privilege for Huawei).
6. Set **VSA Multiple Value Support** to "true" to support multiple values.
7. Specify the **VSA Multiple Value Delimiter**, e.g. ";". The response will contain an attribute like this:
Unknown-29 (29), Length: 5, Data: [7;6], 0x373B36
8. Save the changes.

Example 2:

There are cases where a system user in AccessMatrix server belongs to a certain group, e.g. "GroupA". The VPN vendor that connects to AccessMatrix server does not know about "GroupA", however, this vendor has an internal group (e.g. GroupB) and only users belonging to "GroupB" can access certain information. In this case, the configuration of VSA is required to translate this AccessMatrix group to VPN Vendor group. For example, to return "Fortinet-Group-name" upon authentication, the following vendor-specific attributes should be configured:

1. Set **Enable Vendor-Specific Attributes** to "true" to enable the feature.
2. Select "GroupMembership" as the **VSA Response Class** value. See image below for the User Response sample configuration.

You are here: Configuration > Messaging & Response > User Response

View User Response

*User Response Id:	GroupMembership	*User Response Name:	GroupMembership
Description:		Version :	0

Implementation: User Information Response Collector

User

Fields

Name (Claim Name)	Attribute Name	Delimiter
memberOf	GroupMembership	,

3. Set the Vendor Id to "12356" (Fortinet Vendor ID).
4. Set the key-value pair to **Vendor-Specific Attributes Mapping**. Example:

Vendor Specific Attributes Mapping

Up
Down
Remove

Key	Value	Add
GroupA	GroupB	

GroupA is the AccessMatrix group ID and **GroupB** is the Fortinet group ID.

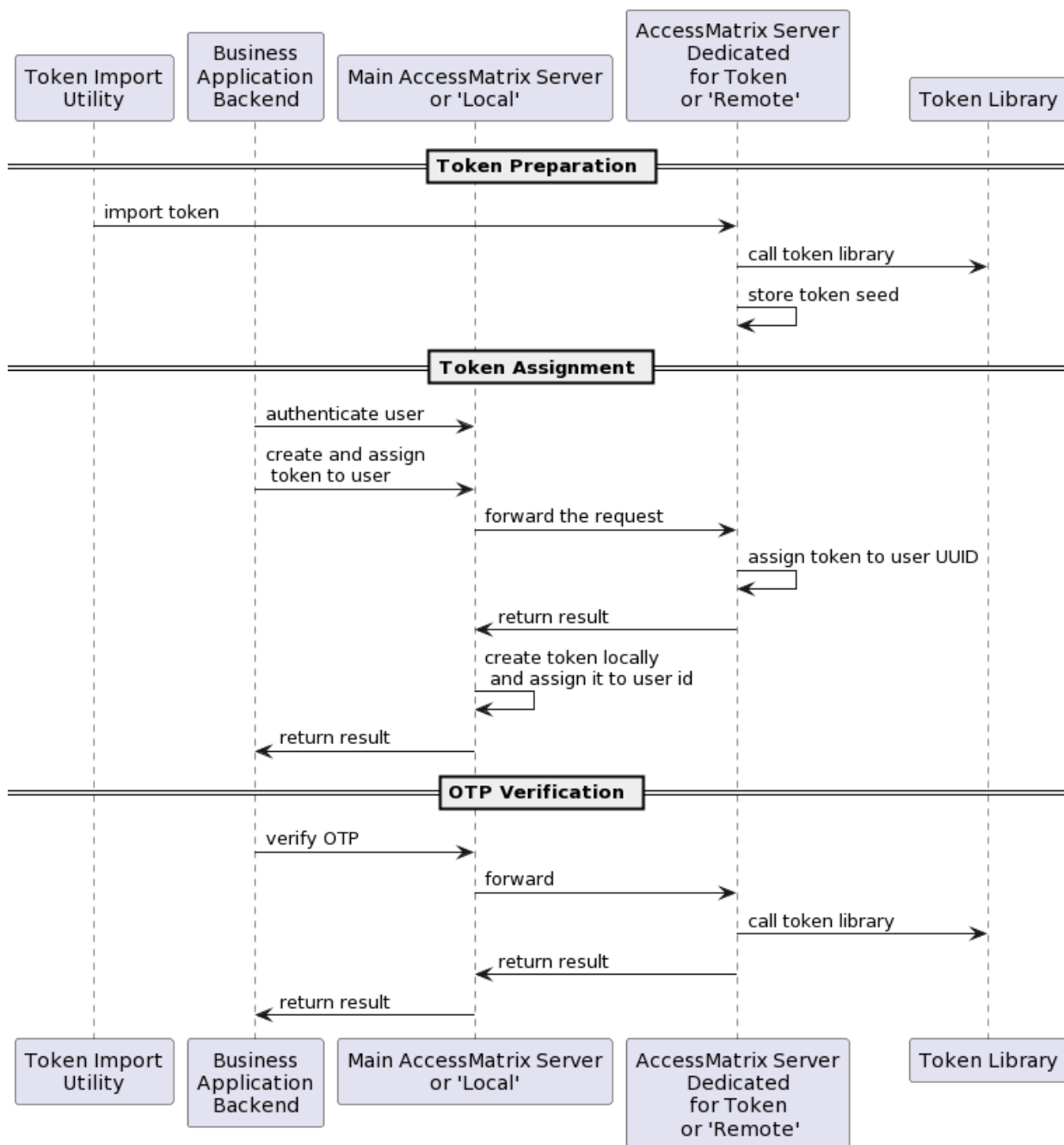
5. Set the **VSA Attribute Tag** to "1" for Fortinet-Group-Name. (Refer to [Fortinet VSA](#).)
 6. Set **VSA Multiple Value Support** to "true" to support multiple values.
 7. Specify the **VSA Multiple Value Delimiter** to ";".
-

-
8. Save the changes.

Vendor Specific Attributes(VSA) 	
Enable Vendor Specific Attributes	true
VSA Response Class	<u>GroupMembership</u>
Vendor Id	12356
Vendor Specific Attributes Mapping	GroupA=GroupB
VSA Attribute Tag	1
VSA Multiple Value Support	true
VSA Multiple Value Delimiter	; (default)

8. Implement Plugin for Remote Token Server

This plugin allows creating and assigning tokens to be located in another AccessMatrix server (remote). In the scenario where a customer deploys a few AccessMatrix servers and would like to segregate the role of each AccessMatrix server. One AccessMatrix server node will be the central authentication server for the application backend. The other AccessMatrix node will be dedicated as a token server. See the simplified concept of these role segregations below.



Note: UUID denotes Universally Unique Identifier.

The tokens' seed is stored in the remote AccessMatrix server. The backend applications identify only the local AccessMatrix server. The API call from the backend application as related to token (creating, assigning, verify OTP and others), will be proxy-ed to the remote AccessMatrix server, transparent to the backend application.

The local AccessMatrix server stores user-token relationship by using user id. The remote AccessMatrix server stores the user-token relationship by using the user's UUID (and the dummy user's store).

When do you use this plugin?

Use this plugin whenever you need to implement a separation of concerns, where one of the AccessMatrix servers will be dedicated as a token server. It might be in a scenario where the customer wants to implement three AccessMatrix servers for SSO for three countries respectively and to centralize the token management in one server.

Where to get the plugin?

It is bundled with the AccessMatrix UAS application. The plugin can be found in the <ACMATRIX_ROOT> folder at C:\<ACMATRIX_ROOT>\sdk\remotetoken\.

How to use the plugin?

Follow the procedural steps below to use the plugin for remote token service.

Install Plugin for Remote Token

Install the plugin into each local SSO server(s) at different locations. Locate the following files in the folder C:\<ACMATRIX_ROOT>\sdk\remotetoken\, and copy to <ACMATRIX_ROOT> folder. See below:

- `\sdk\remotetoken\am-remote-token-core.jar` to `\tomcat\webapps\am5\WEB-INF\lib\` folder
- `\sdk\remotetoken\am-remote-token-plugin.jar` to `\tomcat\plugins\` folder

Next, append **field-definitions.xml** and **wpe.properties** compiled in the executable file **am-res.jar**, which can be found in the folder, <ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\lib\ with the files below respectively:

- `\sdk\remotetoken\res\field-definitions-remotetoken.xml`
- `\sdk\remotetoken\res\wpecustom-remotetoken.properties`

Alternatively, if you do not want to append the existing **field-definitions.xml** and **wpe.properties**, do the following:

-
- Rename `lsdk\remotetoken\res\wpecustom-remotetoken.properties` to **wpecustom.properties**, and then copy to the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\`
 - Copy `lsdk\remotetoken\res\field-definitions-remotetoken.xml` to the folder, `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\`

You must enable "TinyService" for the remote token in the **web.xml** file in the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\` by uncommenting the following lines:

XML

```
<!-- Tiny Rest Remote Token Service -->
<--
,com.isprint.am.remotetoken.TinyRemoteTokenService
-->
```

Set up Remote Token Policy in Admin Console

Set up the remote token policy in Admin Console at the local SSO server(s). The token policy contains the Endpoint URL of the remote token server.

1. From the **Administration Dashboard > Security Policy > Token > Token Policy**.
2. Click **Create** and select "Remote Token Policy" from the implementation drop-down list. Click **Next**.
3. Enter the token policy id, name and description.
4. Enter the Endpoint URL of the remote token server and its keystore information if required.
5. Click **Save** to apply the remote token policy.

Create a token batch

Next, create a token batch that consists of token groups (token group Ids). The token batch will be sent to the remote token server for token management.

1. From the **Operational Dashboard > Token > Token Batch**.
 2. Click **Create**.
 3. Enter token batch id, name, description.
 4. Select the token policy as created on the above steps from the drop-down list.
 5. Click **Save**. A token batch is created.
-

Need to customize Remote Token Service?

AccessMatrix remote token server will not publish the Restful API. You can customize your own Remote Token Service by using this plugin. To do this, use one of the following methods:

- Implement your own **com.isprint.am.remotetoken.IRemoteTokenService**, and then register this class as a default remote token service using the plugin.
- Extend or use a composition of class **com.isprint.am.remotetoken.DefaultRemoteTokenService**, and then customize it with partial methods.
- After you implemented your custom remote token service, register the service in your plugin. For example, if your custom remote token service class is "XMLRPCRemoteTokenSvc", register the default remote token service as follows in your plugin:

Java

```
@Override
    public void load(IPluginContext arg0) {
        logger.info("Plugin loaded: " + getId() + " - " +
            getVersion());
        XMLRPCRemoteTokenSvc xmlRemoteTokenSvc = new
            XMLRPCRemoteTokenSvc();

        DefaultRemoteTokenService.setDefaultService(xmlRemoteTokenSvc);
    }
```

About the version

The plugin is in-built to the AccessMatrix UAS release 5.6.5, 2019 and later versions.

9. Appendices

A. Admin Role and Admin Right Definitions

Admin Role	Description
InitialAdminRole	By default, this initial admin role allows administrator to perform all operations in the system.
UserAdminRole	By default, this user admin role allows administrator to perform following operations: • User Administration (except login account time editing) • Group Administration • ESSO Resource and Entitlement Administration • Token Viewing • Admin Privilege Assignment • Segment Administration • UAM Application Administration
TokenAdminRole	By default, this token admin role allows administrator to perform following operations: • Token Administration • Token Batch Administration • Token Group Administration

Apart from the above predefined admin roles, the administrator is also able to define other admin roles containing desired admin rights. The following table provides detailed descriptions for the various system-defined admin rights in terms of the menu option accessible and actions performable:

Resource	Operation	Menu Option Accessible	Actions Performable
Token Management Rights			
TIU	Import Token	Import Token	Admin Right Dependency: Token - Import, View Token Batch - Create, View Token Group - Create, View i) Able to import token ii) Able to list and create Token Batch

Resource	Operation	Menu Option Accessible	Actions Performable
			iii) Able to list and create Token Group
	Import PIN	Import PIN	Admin Right Dependency: Token - Import, View i) Able to import token
	Reassign Token Group	Reassign Token Group	Admin Right Dependency: Token - View, Assign Token Group Membership Token Group - Create, View i) Able to perform token group reassignment ii) Able to view and create Token Group
	Delete Token	Delete Token	Admin Right Dependency: Token - Import, Delete i) Able to delete the token
	List Token Bulk Jobs	List Token Bulk Jobs	Admin Right Dependency: Token - Import, View i) Able to view the bulk jobs details for the importing jobs
	Export Matrix Card	Export Matrix Card	Admin Right Dependency: Token - Import, View Token Batch - Create, View Token Group - Create, View i) Able to export matrix card ii) Able to create & view for token batch and token group

Resource	Operation	Menu Option Accessible	Actions Performable
	Token Import - PIN Mailer function	Token Import - PIN Mailer function	Admin Right Dependency: Token - Import, View i) Able to generate the PIN Mailer
Token Batch	Create, View, Delete, Edit	Operation Dashboard > Token > Token Batch	Admin Right Dependency: Token Batch - Create, View, Delete, Edit i) Able to create, edit, delete & view token batch
Token Group	Create, View, Delete, Edit	Operation Dashboard > Token > Token Group	Admin Right Dependency: Token Group - Create, View, Delete, Edit i) Able to create, edit, delete & view token group
Token	Change Status	Operation Dashboard > Token > Token	Admin Right Dependency: Token - Edit, View i) Able to search, view, edit and change the status of tokens
	Resynchronize	Operation Dashboard > Token > Token	Admin Right Dependency: Token - View, Resynchronize i) Able to search, view and resynchronize tokens
	Reset Static Password	Operation Dashboard > Token > Token	Admin Right Dependency: Token - View, Reset Password i) Able to search, view and reset the password of tokens
	Resynchronize	Operation Dashboard > Token > Token	Admin Right Dependency: Token - View, Resynchronize

Resource	Operation	Menu Option Accessible	Actions Performable
			i) Able to search, view and resynchronize tokens
User Management Rights			
User	Assign / Unassign Token	Operation Dashboard > User & Segment > User	Admin Right Dependency: User - View, Assign Token i) Able to search users and assign / un-assign tokens to them
System Configuration Rights			
Global Setting	Edit Token Management Attribute	Administration Dashboard > Global Setting > Token Management	Admin Right Dependency: Report - View, Token - View, System - View, Edit i) Able to view & edit the global settings. It's Token Management Attributes in this context
Report Rights			
Token & SMS-related reports	View	Real Time: Token Inventory Report Dashboard > Token & SMS > Real Time > Token Inventory	Token Group Report Admin Right Dependency: Report - View, Token - View, Token Group - View Token Batch Report Admin Right Dependency: Report - View, Token - View, Token Batch - View Token Summary Report Admin Right Dependency: Report - View, Report - Token
		Real Time: Token Inventory Detail By Token Group Report Dashboard > Token & SMS > Real Time > Token Inventory Detail By Token Group	Admin Right Dependency: Report - View, Token - View, Token Group - View
		Real Time: Token Assignment Report Dashboard > Token & SMS > Real Time > Token Assignment	Admin Right Dependency: Report - View, Token -View, User - View

Resource	Operation	Menu Option Accessible	Actions Performable
		Real Time: Token Expiry Report Dashboard > Token & SMS > Real Time > Token Expiry	Admin Right Dependency: Report - View, Token - View, Token Group - View, Token Batch - View
		Historical: Token Activities Report Dashboard > Token & SMS > Historical > Token Activities	Admin Right Dependency: Report - View, Token - View
		Historical: SMS OTP Usage Report Dashboard > Token & SMS > Historical > SMS OTP Usage	Admin Right Dependency: Report - View, Token - View

B. Token Import Utility (TIU) Error Types:

Error Type	Description
NO-SCRIPT-DATA	Potentially caused by an AM5 server disk that is full. Remove some files and try again.
FILE-NOT-FOUND	Potentially caused when one of the AM5 server instances (sharing the same server Id) is not able to find the file uploaded by another instance. Potential workarounds include having each AM5 server instance use a unique server Id by configuring <code>container setting = false</code> , or to specify a common working directory for reports under <code>am.workdir</code> in the amsystem.properties file.

C. ISO 3166-1 Alpha-2 Country Codes

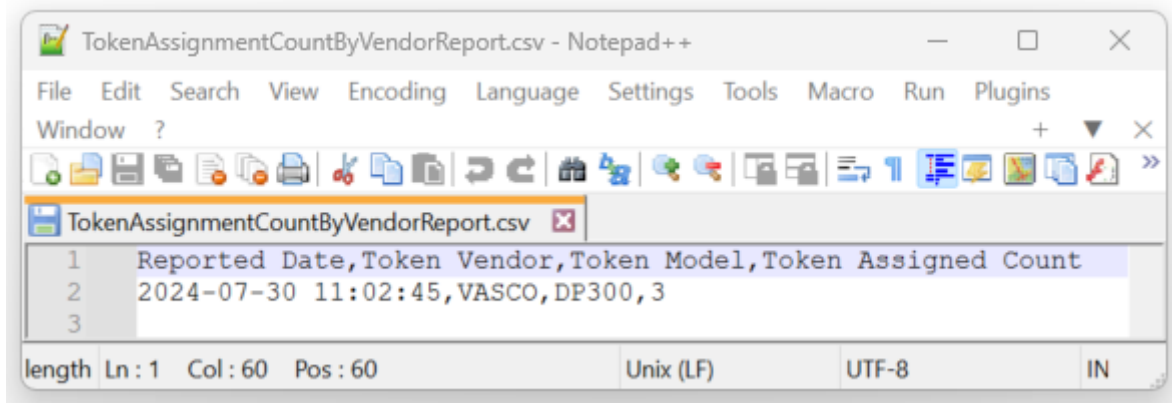
Code	Country name	Code	Country name	Code	Country name	Code	Country name	Code	Country name
AD	Andorra	CU	Cuba	ID	Indonesia	MR	Mauritania	SJ	Svalbard and Jan Mayen
AE	United Arab Emirates	CV	Cabo Verde	IE	Ireland	MS	Montserrat	SK	Slovakia
AF	Afghanistan	CX	Curacao	IL	Israel	MT	Malta	SL	Sierra Leone
AG	Antigua and Barbuda	CW	Christmas Island	IM	Isle of Man	MU	Mauritius	SM	San Marino
AI	Anguilla	CY	Cyprus	IN	India	MV	Maldives	SN	Senegal
AL	Albania	CZ	Czech Republic	IO	British Indian Ocean Territory	MW	Malawi	SO	Somalia
AM	Armenia	DE	Germany	IQ	Iraq	MX	Mexico	SR	Suriname
AO	Angola	DJ	Djibouti	IR	Iran, Islamic Republic of	MY	Malaysia	SS	South Sudan
AQ	Antarctica	DK	Denmark	IS	Iceland	MZ	Mozambique	ST	Sao Tome and Principe
AR	Argentina	DM	Dominica	IT	Italy	NA	Namibia	SV	El Salvador
AS	American Samoa	DO	Dominican Republic	JE	Jersey	NC	New Caledonia	SX	Sint Maarten (Dutch part)
AT	Austria	DZ	Algeria	JM	Jamaica	NE	Niger	SY	Syrian Arab Republic
AU	Australia	EC	Ecuador	JO	Jordan	NF	Norfolk Island	SZ	Swaziland
AW	Aruba	EE	Estonia	JP	Japan	NG	Nigeria	TC	Turks and Caicos Islands
AX	Åland Islands	EG	Egypt	KE	Kenya	NI	Nicaragua	TD	Chad
AZ	Azerbaijan	EH	Western Sahara	KG	Kyrgyzstan	NL	Netherlands	TF	French Southern Territories
BA	Bosnia and Herzegovina	ER	Eritrea	KH	Cambodia	NO	Norway	TG	Togo
BB	Barbados	ES	Spain	KI	Kiribati	NP	Nepal	TH	Thailand
BD	Bangladesh	ET	Ethiopia	KM	Comoros	NR	Nauru	TJ	Tajikistan
BE	Belgium	FI	Finland	KN	Saint Kitts and Nevis	NU	Niue	TK	Tokelau
BG	Burkina Faso	FJ	Fiji	KP	Korea, Democratic People's Republic of	NZ	New Zealand	TL	Timor-Leste
BF	Bulgaria	FK	Falkland Islands (Malvinas)	KR	Korea, Republic of	OM	Oman	TM	Turkmenistan
BH	Bahrain	FM	Micronesia, Federated States of	KW	Kuwait	PA	Panama	TN	Tunisia
BI	Burundi	FO	Faroe Islands	KY	Cayman Islands	PE	Peru	TO	Tonga
BJ	Benin	FR	France	KZ	Kazakhstan	PF	French Polynesia	TR	Turkey
BL	Saint Barthélemy	GA	Gabon	LA	Lao People's Democratic Republic	PG	Papua New Guinea	TT	Trinidad and Tobago
BM	Bermuda	GB	United Kingdom	LB	Lebanon	PH	Philippines	TV	Tuvalu
BN	Brunei Darussalam	GD	Grenada	LC	Saint Lucia	PK	Pakistan	TW	Taiwan, Province of China
BO	Bolivia, Plurinational State of	GE	Georgia	LI	Liechtenstein	PL	Poland	TZ	Tanzania, United Republic of
BQ	Bonaire, Sint Eustatius and Saba	GF	French Guiana	LK	Sri Lanka	PM	Saint Pierre and Miquelon	UA	Ukraine
BR	Brazil	GG	Guernsey	LR	Liberia	PN	Pitcairn	UG	Uganda
BS	Bahamas	GH	Ghana	LS	Lesotho	PR	Puerto Rico	UM	United States Minor Outlying Islands
BT	Bhutan	GI	Gibraltar	LT	Lithuania	PS	Palestine, State of	US	United States
BV	Bouvet Island	GL	Greenland	LU	Luxembourg	PT	Portugal	UY	Uruguay
BW	Botswana	GM	Gambia	LV	Latvia	PW	Palau	UZ	Uzbekistan
BY	Belarus	GN	Guinea	LY	Libya	PY	Paraguay	VA	Holy See (Vatican City State)
BZ	Belize	GQ	Guadeloupe	MA	Morocco	QA	Qatar	VC	Saint Vincent and the Grenadines
CA	Canada	GQ	Equatorial Guinea	MC	Monaco	RE	Réunion	VE	Venezuela, Bolivarian Republic of
CC	Cocos (Keeling) Islands	GR	Greece	MD	Moldova, Republic of	RO	Romania	VG	Virgin Islands, British
CD	Congo, the Democratic Republic of the	GS	South Georgia and the South Sandwich Islands	ME	Montenegro	RS	Serbia	VI	Virgin Islands, U.S.
CF	Central African Republic	GT	Guatemala	MF	Saint Martin (French part)	RU	Russian Federation	VN	Viet Nam
CG	Congo	GU	Guam	MG	Madagascar	RW	Rwanda	VU	Vanuatu
CH	Switzerland	GW	Guinea-Bissau	MH	Marshall Islands	SA	Saudi Arabia	WF	Wallis and Futuna
CI	Côte d'Ivoire	GY	Guyana	MK	Macedonia, the former Yugoslav Republic of	SB	Solomon Islands	WS	Samoa
CK	Cook Islands	HK	Hong Kong	ML	Mali	SC	Seychelles	YE	Yemen
CL	Chile	HM	Heard Island and McDonald Islands	MM	Myanmar	SD	Sudan	YT	Mayotte
CM	Cameroon	HN	Honduras	MN	Mongolia	SE	Sweden	ZA	South Africa
CN	China	HR	Croatia	MO	Macao	SG	Singapore	ZM	Zambia
CO	Colombia	HT	Haiti	MP	Northern Mariana Islands	SH	Saint Helena, Ascension and Tristan da Cunha	ZW	Zimbabwe
CR	Costa Rica	HU	Hungary	MQ	Martinique	SI	Slovenia		

D. UAS Sample Report Templates

The following token report samples are generated using the following dataset:

Quantity	Assigned?	Token Model	Token Batch	Token Group
3	Yes	DP300	TB_01	TG_01
2	No	DP300	TB_01	TG_01
1	No	DP275	TB_02	TG_02

Token Assignment Count By Vendor Report Sample



1	Reported Date,Token Vendor,Token Model,Token Assigned Count
2	2024-07-30 11:02:45,VASCO,DP300,3
3	

length Ln : 1 Col : 60 Pos : 60 Unix (LF) UTF-8 IN

Token Inventory Report Sample

By Summary

Sample report with **Token Group Filter Mode** criteria set to "None":



Token Inventory By Summary Report

Report Filtering Criteria

Token Model	: All	Include Pie Chart	: No
Token Status Display List	: Filter by token status display list in global setting		
Token Group Filter Mode	: None		
Token Group Id	: All		

Token Vendor : VASCO

Token Model : DP300

Token Status	Token Assigned	Token Unassigned	Total Number
Activated	3	1	4
Battery Expired	0	0	0
Damaged	0	1	1
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsusended	0	0	0
Total Number	3	2	5



Token Inventory By Summary Report

Token Vendor	: VASCO	Token Model	: DP275
Token Status	Token Assigned	Token Unassigned	Total Number
Activated	0	1	1
Battery Expired	0	0	0
Damaged	0	0	0
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unuspended	0	0	0
Total Number	0	1	1

Sample report with **Token Group Filter Mode** criteria set to "Include", with **Token Model** criteria specified:



Token Inventory By Summary Report

Report Filtering Criteria			
Token Model	: DP300	Include Pie Chart	: No
Token Status Display List	: Filter by token status display list in global setting		
Token Group Filter Mode	: Include		
Token Group Id	: TG_01		

Token Vendor	: VASCO	Token Model	: DP300
Token Status	Token Assigned	Token Unassigned	Total Number
Activated	3	1	4
Battery Expired	0	0	0
Damaged	0	1	1
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspending	0	0	0
Total Number	3	2	5

Sample report with **Token Group Filter Mode** criteria set to "Exclude", with **Token Model** criteria specified:



Token Inventory By Summary Report

Report Filtering Criteria			
Token Model	: DP275	Include Pie Chart	: No
Token Status Display List	: Filter by token status display list in global setting		
Token Group Filter Mode	: Exclude		
Token Group Id	: TG_01		

Token Vendor	: VASCO	Token Model	: DP275
Token Status	Token Assigned	Token Unassigned	Total Number
Activated	0	1	1
Battery Expired	0	0	0
Damaged	0	0	0
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspended	0	0	0
Total Number	0	1	1

By Batch

Printed Date : 2024-07-30 11:06:30

Time zone : SGT

Page : 1 / 1



Token Inventory By Batch Report

Report Filtering Criteria

Token Batch Id : TB_01

Token Model : All

Token Status Display List : Filter by token status display list in global setting

Date Range of Token Batch Manufacture Time : -

Token Batch To Be Expired Within (No. Of Months) : - Include Custom Attribute : No

Expiry Time : 2024-08-15 00:00:00

Manufacture Time : -

Token Batch Id : TB_01

Token Vendor : VASCO

Token Batch Name : TB_01

Token Model : DP300

Token Policy : DefaultVascoKernelParameters

Static Vector : -

Token Status	Token Assigned	Token Unassigned	Total Number
Activated	3	1	4
Battery Expired	0	0	0
Damaged	0	1	1
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspending	0	0	0
Total Number	3	2	5

By Group

Sample report with **Include Token Model Filtering** criteria set to "No":



Token Inventory By Group Report

Report Filtering Criteria

Token Group Id : All

Include Token Model Filtering : No

Token Model : -

Token Status Display List : Filter by token status display list in global setting

Token Group Id : TG_01

Token Group Name : TG_01

Token Vendor : VASCO

Token Status	Token Assigned	Token Unassigned	Total Number
Activated	3	1	4
Battery Expired	0	0	0
Damaged	0	1	1
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspending	0	0	0
Total Number	3	2	5



Token Inventory By Group Report

Token Group Id : TG_02

Token Group Name : TG_02

Token Vendor : VASCO

Token Status	Token Assigned	Token Unassigned	Total Number
Activated	0	1	1
Battery Expired	0	0	0
Damaged	0	0	0
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspended	0	0	0
Total Number	0	1	1

Sample report with **Include Token Model Filtering** criteria set to "Yes", with **Token Model** criteria specified:



Token Inventory By Group Report

Report Filtering Criteria

Token Group Id : All

Include Token Model Filtering : Yes

Token Model : DP300

Token Status Display List : Filter by token status display list in global setting

Token Group Id : TG_01

Token Group Name : TG_01

Token Vendor : VASCO

Token Model : DP300

Token Status	Token Assigned	Token Unassigned	Total Number
Activated	3	1	4
Battery Expired	0	0	0
Damaged	0	1	1
Deactivated	0	0	0
Deleted	0	0	0
Elevated	0	0	0
Expired	0	0	0
Faulty	0	0	0
Frozen	0	0	0
In_Transit	0	0	0
Invalidated	0	0	0
Locked	0	0	0
Lost	0	0	0
PreActivation	0	0	0
Returned	0	0	0
Suspended	0	0	0
Unassigned	0	0	0
Unsuspending	0	0	0
Total Number	3	2	5

Token Inventory Detail By Token Group Report Sample

Printed Date : 2024-07-30 11:23:44

Time zone : SGT

Page : 1 / 1

Token Inventory Detail By Token Group Report

Report Filtering Criteria			
From Token Serial Number	: -	To Token Serial Number	: -
Token Group Id	: TG_01	Token Model	: -
Token Status	: -		
Sorted Order	: Token Serial Number		

Token Group Id : TG_01						
Token Serial Number	Token Vendor	Token Model	Token Status	Assignee Id@User Store	Assignee Name	Assigned Time
0097000001	VASCO	DP300	Activated	demouser@DefaultStore	Demo User	2024-07-29 14:15:00
0097000002	VASCO	DP300	Activated	testuser@DefaultStore	Test User	2024-07-29 14:15:32
0097000003	VASCO	DP300	Activated	demouser2@DefaultStore	Demo User 2	2024-07-30 10:02:03
0097000004	VASCO	DP300	Activated	-	-	-
0097000005	VASCO	DP300	Damaged	-	-	-

- End Of Report -

Token Assignment Report Sample



Report Filtering Criteria			
From Token Serial Number	: -	To Token Serial Number	: -
From Date	: 2024-07-14 00:00:00	To Date	: 2024-07-30 11:24:44
Sorted Order	: Assignee Id		

Assignee Id/Token Sharing Group ID@User Store	Assignee Name/ (User Store)Assignee Id	Token Serial Number	Token Vendor	Token Model	Token Status	Assigned Time	Assigner Id@User Store
demouser2@DefaultStore	Demo User 2/-	0097000003	VASCO	DP300	Activated	2024-07-30 10:02:03	amadmin@DefaultStore
demouser@DefaultStore	Demo User/-	0097000001	VASCO	DP300	Activated	2024-07-29 14:15:00	amadmin@DefaultStore
testuser@DefaultStore	Test User/-	0097000002	VASCO	DP300	Activated	2024-07-29 14:15:32	amadmin@DefaultStore

Total Assigned : 3

- End Of Report -

Token Expiry Report Sample

Printed Date : 2024-07-30 11:27:20

Time zone : SGT

Page : 1 / 2



Token Expiry Report

Report Filtering Criteria

Token Group Id : TG_01

Token Batch Id : All

Token Status : -

Include Custom Attribute : No

Token Model : All

Assignment State : -

Date Range of Token Batch Manufacture Time : -

Token Batch To Be Expired Within (No. Of Months) : -

Expiry Time : 2024-08-15 00:00:00

Manufacture Time : -

Token Batch Id : TB_01

Token Vendor : VASCO

Token Batch Name : TB_01

Token Model : DP300

Token Policy : DefaultVascoKernelParameters

Token Group Id : TG_01

Token Status : Activated

Assignment State : Assigned

Token Serial Number From : 0097000001

Token Serial Number To : 0097000003

Static Vector : -

Token Serial Number

0097000001

0097000002

0097000003

Expiry Time : 2024-08-15 00:00:00

Manufacture Time : -

Token Batch Id : TB_01

Token Vendor : VASCO

Token Batch Name : TB_01

Token Model : DP300

Token Policy : DefaultVascoKernelParameters

Token Group Id : TG_01

Token Status : Activated

Assignment State : Unassigned

Token Serial Number From : 0097000004

Token Serial Number To : 0097000004

Static Vector : -

Token Serial Number

0097000004

Token Activities Report Sample

Printed Date : 2024-07-30 11:28:40

Time zone : SGT

Page : 1 / 2



Token Activities Report

Report Filtering Criteria

User Store	: All	Event Result	: All
Actor User Id	: All	Include actions done by background tasks	: No
From Token Serial Number	: -	To Token Serial Number	: -
Event Start	: 2024-07-30 00:00:00	Event End	: 2024-07-30 11:27:27
Agent User Id	: All	User Store	: All
Target Id	: All	Token Vendor	: -
Query Limit	: 1000	Is Query Result Exceeding Query Limit	: false
Event	: All		

Audit Id (Server Id) / Actor User Id@User Store	Event Timestamp	Event	Batch/Group/Token/User	Token Serial Number(s)/Digital Signature Transaction Data	Token Count/ Token Status/ Token Sharing Group ID	Event Result / Message/ Result Code	Device Friendly Name/ Device Instance Number/ Device Type
440(Aloysius-T14s)/ amadmin@DefaultStore	2024-07-30 10:00:37	Unassign	User (demouser)	OATH (GA20240718173951543)-	-/ Unassigned/ -	Successful	-/-
441(Aloysius-T14s)/ amadmin@DefaultStore	2024-07-30 10:01:00	Delete	Token (OATH-GA20240718173951543)	-/-	-/-	Successful	-/-
442(Aloysius-T14s)/ amadmin@DefaultStore	2024-07-30 10:01:08	Delete	Token (VASCO-1097123456)	-/-	-/-	Successful	-/-
444(Aloysius-T14s)/ amadmin@DefaultStore	2024-07-30 10:02:03	Assign	User (demouser2)	VASCO (0097000003)-	-/ Activated/ -	Successful	-/-

- 1 -

SMS OTP Usage Report Sample

Printed Date : 2017-05-24 14:34:22

Time zone : SGT

Page : 1 / 1



SMS OTP Usage Report

Report Filtering Criteria

OTP Request Frequency	: > 1	Start Date	: 2017-05-24 00:00:00
Actor Segment Store	: All	Actor Segment Id	:
End Date	: 2017-05-24 14:34:19	Sorted Order	: User Id

No.	Actor User Id@ User Store	User Segment Id@ Segment Store	User name	Mobile Number	Email	No. Of SMS
1	user1@ DefaultStore	-	user1	-	joy.lumancas@i-sprint.com	2
2	user2@ DefaultStore	-	user2	-	joy.lumancas@i-sprint.com	2
